# A LIGHTWEIGHT SPARSE MIXTURE-OF-EXPERTS FRAMEWORK FOR INTRUSION DETECTION IN RESOURCE-CONSTRAINED IOT NETWORKS

BY

UDOH, BENJAMIN OKON

REGISTRATION NUMBER: 24/PG/CE/SC/MCS/004

SUPERVISED BY

DR. BLISS U. STEPHEN

TETFUND CENTRE OF EXCELLENCE IN COMPUTATIONAL INTELLIGENCE RESEARCH, UNIVERSITY OF UYO.

AUGUST 2026

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A SEMINAR SUBMITTED TO THE TETFUND CENTRE OF EXCELLENCE IN COMPUTATIONAL INTELLIGENCE RESEARCH, UNIVERSITY OF UYO, UYO, AKWA IBOM STATE, NIGERIA, IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF A MASTER'S DEGREE.

SUPERVISED BY

DR. BLISS U. STEPHEN

AUGUST 2026

# DECLARATION

I, UDOH, BENJAMIN OKON, hereby declare that this seminar entitled "A Lightweight Sparse Mixture-of-Experts Framework for Intrusion Detection in Resource-Constrained IoT Networks" was written by me and that it is the correct record of my own research work. It has not been presented for a degree in another institution. I will not present it, or cause it to be presented, for a degree in another institution. All sources of information have been appropriately acknowledged using references and other acceptable methods.

Signed: . . . . . Date: . . . . . .

UDOH, BENJAMIN OKON

# CERTIFICATION

This seminar entitled "A Lightweight Sparse Mixture-of-Experts Framework for Intrusion Detection in Resource-Constrained IoT Networks" by UDOH, BENJAMIN OKON (24/PG/CE/SC/MCS/004) meets the regulations governing the award of a Master's degree of the University of Uyo. The work has made original contribution to knowledge. It should be submitted to the Postgraduate School for approval.

(a) Supervisor

Signature: . . . . . Date: . . . . . .

Name: . . . . . Rank: . . . . . .

Department/Faculty: . . . . . .

(b) Co-Supervisor, if any

Signature: . . . . . Date: . . . . . .

Name: . . . . . Rank: . . . . . .

Department/Faculty: . . . . . .

(c) Internal Examiner

Signature: . . . . . Date: . . . . . .

Name: . . . . . Rank: . . . . . .

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Name: . . . . . Rank: . . . . . .

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# DEDICATION

I dedicate this work to God Almighty, who never fails to protect, guide and give break through financially throughout the course of this programme. To my dear wife, Dr. Idaresit Udoh for her support, patience and encouragement. To my children Wilfred, Jahswill and Princewill, who are always showering me with prayers and a source of motivation to me and always a reason to smile. To my siblings and friends, for their continued support and encouragement throughout this journey.

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I am thankful to everyone who has contributed, in their own unique way, to the completion of this thesis. Thank you, and God bless.

# ABSTRACT

The rapid proliferation of Internet of Things (IoT) devices has widened the attack surface of modern networks, yet the intrusion detection systems that protect them must operate within severe processing, memory and energy constraints. Deep and transformer based detectors achieve high accuracy but are frequently too heavy, and often too opaque, for deployment on constrained IoT hardware. This study designed, implemented and evaluated a lightweight sparse Mixture-of-Experts (MoE) framework that reconciles detection accuracy, computational efficiency and interpretability for resource-constrained IoT intrusion detection. The framework couples a shared feature embedding with a pool of eight architecturally heterogeneous, individually lightweight experts, a noisy top-$k$ router regularised by a load balancing objective, and a fusion classification head, so that only a small subset of experts is evaluated for each network flow. Two heterogeneous public benchmark corpora, IoT-23 and Edge-IIoTset, were processed through a leakage-aware pipeline comprising the removal of uninformative, identifying and target-leaking attributes, per class Tukey winsorisation, stratified partitioning, training-partition-only scaling and minority-only synthetic oversampling, and the framework was evaluated against seven baseline detectors spanning classical, deep, hybrid and transformer families in both binary and multiclass regimes. The proposed framework matched or nearly matched the strongest baseline in every configuration, attaining 99.88% accuracy on IoT-23 binary detection, 99.87% macro F1 on IoT-23 multiclass detection, and 99.84% accuracy with 95.55% macro F1 on the demanding twelve-class Edge-IIoTset task, and it was the only neural model to remain consistent across all configurations. Efficiency analysis confirmed that its active per-prediction cost, rather than its nominal capacity, is a fraction of that of a dense model of equal capacity, while a four-level explainability analysis and an architectural ablation established that its routing is semantically meaningful and that heterogeneity, moderate sparsity and pool size each contribute. The study concludes that conditional computation over a heterogeneous expert pool is a practical route to accurate, efficient and interpretable intrusion detection on the constrained IoT edge, and recommends its extension to federated and on-device deployment.

[325 words]

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# CHAPTER ONE

# INTRODUCTION

## 1.1. Background of the Study

The Internet of Things (IoT) has evolved from a specialised networking idea into a core model of modern computation. that underpins modern digital infrastructure, connecting actuators, sensors, and embedded controllers across homes, industries, cities, farms, and health-care facilities (Dritsas & Trigka, 2025; Mishra & Mishra, 2025). The number and diversity of connected devices continue to expand rapidly, and much of this growth is concentrated in resource-constrained nodes that operate under strict limitations of processing power, memory, storage, and battery energy (Altaibek *et al.*, 2025; Sharma & Bhushan, 2026). This growth has been mirrored by the emergence of large-scale deployment domains such as Industrial IoT (IIoT), smart cities, smart homes, and health-care IoT, each of which introduces its own operational and security requirements (Fraihat *et al.*, 2026; Khatun *et al.*, 2023).

The convenience and automation offered by IoT are, however, accompanied by an expanding attack surface. Because IoT devices are frequently deployed with limited hardware capacity and heterogeneous firmware, they are attractive targets for adversaries seeking to exploit weak authentication, unpatched vulnerabilities, and insecure communication protocols (Almutairi & Sheldon, 2025; Waqdan *et al.*, 2025). Cyber threats directed at IoT ecosystems range from botnets and distributed denial-of-service (DDoS) campaigns to reconnaissance, spoofing, and increasingly sophisticated zero-day exploits that evade static defences (Altaibek *et al.*, 2025; Hssayeni & Mahgoub, 2026). Consequently, Intrusion Detection Systems (IDS) have become a central pillar of IoT security, complementing cryptographic and access-control mechanisms by continuously monitoring network behaviour for malicious activity (Dritsas & Trigka, 2025; Sharma & Bhushan, 2026).

Traditional signature-based IDS approaches, while effective against known attack patterns, are structurally unable to detect novel or zero-day attacks because they depend on pre-defined

attack signatures (Hssayeni & Mahgoub, 2026). This limitation motivated a shift towards anomaly-based detection using machine learning (ML), where classifiers such as random forest, support vector machines, and gradient-boosted trees are trained to distinguish benign from malicious traffic (Sadhwani *et al.*, 2024; Verma & Ranga, 2020). More recently, Deep Learning (DL) architectures, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and their hybrid combinations, have demonstrated strong capability in automatically learning discriminative spatial and temporal representations from raw or lightly processed network traffic (Agbor *et al.*, 2025; Altunay & Albayrak, 2023; Gueriani *et al.*, 2024).

As detection architectures have grown more capable, they have also grown more computationally demanding. Transformer-based and vision-transformer-inspired IDS models have achieved impressive detection accuracy by capturing long-range dependencies in traffic sequences (Almadhor *et al.*, 2025; C. Du *et al.*, 2024; Sana *et al.*, 2024), and pre-trained language models have been fine-tuned for threat classification and threat-pattern discovery in industrial settings (Liu *et al.*, 2025; Ma *et al.*, 2025). Graph Neural Networks (GNN) and graph attention mechanisms have likewise been explored to model the relational structure of network flows, often in self-supervised settings that reduce dependence on labelled data (H. Nguyen & Kashef, 2023; R. Xu *et al.*, 2024). In parallel, Federated Learning (FL) has been adopted to enable collaborative, privacy-preserving model training across distributed IoT clients without centralising raw traffic data (Belarbi *et al.*, 2023; Chatterjee & Hanawal, 2022; T. D. Nguyen *et al.*, 2026).

Despite these advances, the growing computational and memory footprint of state-of-the-art detectors, particularly deep transformer and large-language-model-based approaches, limits their direct deployment on constrained IoT endpoints and gateways (Adjewa *et al.*, 2026; Almadhor *et al.*, 2025). This has motivated a distinct research thread on lightweight and efficient AI for edge deployment, encompassing model compression, knowledge distillation, quantisation, and compact hybrid architectures (Al-Nomasy *et al.*, 2025; Jouhari & Guizani, 2024; Z. Li & Yao, 2024). Sparse Mixture-of-Experts (MoE) architectures have emerged in the broader machine-learning literature as an efficient alternative to dense models: by routing each input to only a small subset of specialised expert sub-networks, sparse MoE models can, in principle, decouple model capacity from per-sample computational cost (Ilias *et al.*, 2026). Although MoE has

begun to appear in federated intrusion-detection research (Ilias *et al.*, 2026), its application to lightweight, deployment-aware IoT intrusion detection remains limited, motivating the present study.

### 1.2. Statement of the Research Problem

The rapid expansion of IoT ecosystems has widened the attack surface available to adversaries, while the devices that make up the IoT ecosystems typically possess limited CPU, memory, storage, and battery capacity (Altaibek *et al.*, 2025; Sharma & Bhushan, 2026). Cyberattacks against IoT networks are also becoming more sophisticated, encompassing zero-day exploits, multi-stage advanced persistent threats, and adversarially adaptive traffic patterns that are difficult for static, signature-based systems to detect (Hssayeni & Mahgoub, 2026; Sundaravadivel *et al.*, 2026). Signature-based IDS by design are inherently unable to detect previously unseen attacks, and many anomaly-based ML approaches still exhibit unacceptably high false-alarm rates in operational settings (Sadhwani *et al.*, 2024; Verma & Ranga, 2020).

While deep learning and, more recently, transformer-based and large-language-model-based IDS have improved intrusion detection performance considerably (C. Du *et al.*, 2024; Ma *et al.*, 2025; Sana *et al.*, 2024), these gains have generally come at the cost of heavier computation, larger parameter counts, and increased memory usage, all of which are poorly matched to the resource envelope of IoT hardware (Adjewa *et al.*, 2026; Almadhor *et al.*, 2025). Ensemble and hybrid deep-learning models further improve accuracy by combining multiple learners but typically increase architectural complexity and inference latency rather than reducing it (Agbor *et al.*, 2025; Lazzarini *et al.*, 2023; Wakili & Bakkali, 2025). Federated learning mitigates privacy and centralisation concerns but introduces its own challenges of client heterogeneity, non-independent and identically distributed (non-IID) data, and communication overhead (Belarbi *et al.*, 2023; Ilias *et al.*, 2026; T. D. Nguyen *et al.*, 2026).

Sparse Mixture-of-Experts architectures offer a potential route out of this trade-off by activating only a fraction of available model capacity for each input, but their application to IoT intrusion detection has, to date, been largely restricted to a small number of federated learning studies (Ilias *et al.*, 2026), with limited attention to lightweight expert design, adaptive

or resource-aware routing, and joint evaluation of detection accuracy alongside computational efficiency metrics such as latency, memory footprint, and FLOPs. In addition, explainability is rarely integrated directly into advanced IDS architectures, even though transparency is increasingly recognised as important for operator trust and root-cause analysis (Al-Nomasy *et al.*, 2025; Attique *et al.*, 2024; Wakili & Bakkali, 2025).

Therefore, there is a need for an intelligent, lightweight, scalable, and computationally efficient intrusion detection framework capable of maintaining high detection performance while remaining suitable for deployment on resource-constrained devices in IoT ecosystems.

### 1.3. Objectives of the Study

The main objective of this study is to develop a lightweight sparse mixture-of-experts based intrusion detection framework that balances accuracy, interpretability and computational efficiency for resource-constrained iot environments.

The specific objectives include the following:

1. To collect and characterise the IoT-23 and Edge-IIoTset datasets for binary and multi class intrusion detection.

2. To develop a data preprocessing pipeline that is leakage-aware and a lightweight sparse mixture-of-experts architecture to achieve efficient intrusion detection.

3. To evaluate the proposed framework against baseline models across both datasets and detection settings using performance and efficiency metrics.

4. To evaluate model interpretability through expert routing analysis, global feature attribution and local prediction explanations.

### 1.4. Significance of the Study

This study is expected to contribute to several interrelated domains. From a security standpoint, it advances the state of the art in IoT security by proposing a detection framework tailored to the resource realities of constrained devices (Altaibek *et al.*, 2025; Sharma & Bhushan, 2026). Its

relevance extends to Industrial IoT and critical infrastructure protection, where reliable and low-latency intrusion detection is essential to safeguarding operational technology (AL Ibaisi *et al.*, 2026; Fraihat *et al.*, 2026). In smart-city and smart-home contexts, the proposed lightweight architecture can support on-device or near-edge detection without reliance on constant cloud connectivity (Almutairi & Sheldon, 2025). In health-care IoT, where both privacy and latency are critical, an efficient and explainable detector contributes to safer deployment of connected medical devices (Khatun *et al.*, 2023).

From an AI-based cybersecurity and edge-computing perspective, the study contributes new evidence on how sparse, conditional-computation architectures can be adapted for security-relevant tasks beyond their traditional natural-language-processing origins (Ilias *et al.*, 2026). It also contributes to deep learning research more broadly by demonstrating heterogeneous expert design and lightweight routing strategies. Its explainable-AI component adds to the growing body of work that seeks to make advanced IDS architectures interpretable to human analysts (Al-Hejri *et al.*, 2026; Al-Nomasy *et al.*, 2025; Wakili & Bakkali, 2025). Finally, by explicitly reporting efficiency metrics alongside detection metrics, the study offers a template that may inform future IoT security standards and evaluation practice.

### 1.5. Scope of the Study

#### Datasets

The scope of the study includes sourcing publicly available IoT and IIoT network-traffic intrusion-detection datasets that are widely used in recent literature, including CICIoT2023, Edge-IIoTset and IoT-23 as applicable to the experimental design (Agbor *et al.*, 2025; AL Ibaisi *et al.*, 2026; Altunay & Albayrak, 2023; Gueriani *et al.*, 2024; Sadhwani *et al.*, 2024).

#### Models

The scope covers baseline machine learning models, CNN, LSTM, and transformer-based deep learning models, hybrid and ensemble models drawn from the literature; and the proposed sparse Mixture-of-Experts model (Altunay & Albayrak, 2023; Ilias *et al.*, 2026; Lazzarini *et al.*, 2023; Sana *et al.*, 2024; Verma & Ranga, 2020).

**Evaluation Metrics**

Evaluation is scoped to standard classification metrics (accuracy, precision, recall, F1-score, ROC-AUC, false-positive rate, detection rate) and efficiency metrics (inference time, FLOPs, parameter count, memory usage, and, where measurable, energy consumption).

## 1.6. Limitations of the Study

The study is subject to the following limitations:

1. The evaluation relies on publicly available benchmark datasets rather than live production traffic.

2. Attack scenarios are drawn from simulated or previously captured traffic rather than real-time adversarial campaigns.

3. Hardware-based evaluation is beyond the scope of this study due to the limited availability of IoT and edge computing devices. Therefore, the performance of the proposed model may not generalise across all IoT and edge hardware platforms.

4. Experimental evaluation is conducted on individual datasets rather than through cross-dataset or domain-adaptive evaluation. Consequently, the robustness and generalisability of the proposed model across diverse data distributions and deployment environments are not assessed.

5. No live, in-production deployment of the proposed framework is undertaken within the scope of this study.

## 1.7. Definition of Key Terms

**Internet of Things (IoT)**

A network of interconnected physical devices embedded with sensors, software, and communication capabilities that allow them to collect and exchange data (Mishra & Mishra, 2025).

**Resource-Constrained Devices**

IoT devices characterised by limited processing power, memory, storage, and energy budgets, which restrict the complexity of models that can be deployed on them (Altaibek *et al.*, 2025).

**Intrusion Detection System (IDS)**

A system that monitors network or host activity to detect malicious behaviour or policy violations (Sharma & Bhushan, 2026).

**Edge Computing**

A computing paradigm in which data processing is performed close to the data source rather than in a centralised cloud, reducing latency and bandwidth requirements.

**TinyML**

The practice of designing and deploying machine learning models capable of running on microcontrollers and other severely resource-constrained hardware.

**Deep Learning**

A subfield of machine learning based on multi-layered artificial neural networks capable of learning hierarchical representations from data.

**Federated Learning**

A distributed machine learning paradigm in which multiple clients collaboratively train a shared model without transmitting their raw local data to a central server (Belarbi *et al.*, 2023).

**Explainable AI (XAI)**

A set of methods and techniques that make the predictions of machine learning models interpretable and understandable to humans (Wakili & Bakkali, 2025).

**Sparse Mixture-of-Experts (MoE)**

A neural network architecture composed of multiple expert sub-networks and a gating/routing mechanism that activates only a subset of experts for each input, decoupling model capacity from per-sample computation (Ilias *et al.*, 2026).

**Transformer**

A neural network architecture based on self-attention mechanisms capable of modelling long-range dependencies in sequential data (Sana *et al.*, 2024).

**Graph Neural Network (GNN)**

A class of neural networks designed to operate directly on graph-structured data by propagating and aggregating information across nodes and edges (R. Xu *et al.*, 2024).

Continue ....

# CHAPTER TWO

# LITERATURE REVIEW

## 2.1. Preamble

This chapter reviews the literature relevant to intrusion detection in resource-constrained internet of things (IoT) networks. It begins with the foundational concepts of the IoT and its security requirements before progressing through the major families of intrusion-detection techniques that have been applied in this domain: machine learning, deep learning, ensemble learning, hybrid deep learning, graph-based learning, transformer-based models, lightweight and edge-oriented AI, federated learning, explainable AI, and, finally, sparse Mixture-of-Experts architectures. The chapter also reviews commonly used public datasets and evaluation metrics, synthesises recent related studies published between 2023 and 2026, and concludes by articulating the research gaps that motivate the present study.

## 2.2. Internet of Things (IoT)

Internet of Things (IoT) refers to a network of physical objects, ranging from sensors and actuators to household appliances and industrial machinery, that are embedded with computing and communication capabilities enabling them to sense, exchange, and act upon data with minimal human intervention (Mishra & Mishra, 2025; Sharma & Bhushan, 2026). Characteristic features of IoT systems include heterogeneity of hardware and communication protocols, large-scale connectivity, and, in many deployments, severe resource constraints in terms of processing power, memory, storage, and energy (Altaibek *et al.*, 2025). Architecturally, IoT systems are commonly described in terms of perception, network, and application layers, with intermediate fog or edge layers increasingly interposed to reduce latency and offload computation from constrained end devices (Almutairi & Sheldon, 2025; Khatun *et al.*, 2023). Application domains include smart cities, smart homes, industrial automation, agriculture, and health care, each imposing different security, latency, and reliability requirements (Khatun *et al.*, 2023; Sundaravadivel *et al.*, 2026).

Resource-constrained devices, such as microcontrollers, sensor nodes, and embedded gateways, form the backbone of many of these deployments and are simultaneously the most difficult to secure, since conventional security software is often too heavy for their hardware (Altaibek *et al.*, 2025).

## 2.3. IoT Security

### 2.3.1. Security Requirements

IoT security is generally framed around the classical information-security requirements of confidentiality, integrity, and availability, extended to account for the constrained and distributed nature of IoT deployments (Mishra & Mishra, 2025; Waqdan *et al.*, 2025). Authentication and authorisation mechanisms must verify the identity of devices and users while remaining lightweight enough for constrained hardware (Mishra & Mishra, 2025). Privacy is an additional and increasingly prominent requirement, particularly in domains such as health-care IoT, where sensitive personal data is continuously generated and transmitted (Khatun *et al.*, 2023). The integration of IoT with cloud and edge infrastructures introduces further requirements around secure multi-tenant isolation and end-to-end security architectures spanning both device and infrastructure layers (Almutairi & Sheldon, 2025).

### 2.3.2. IoT Threat Landscape

The IoT threat landscape encompasses a broad range of attack types, including malware infections, botnet recruitment, distributed denial-of-service (DDoS) attacks, ransomware, spoofing, Sybil attacks, routing attacks, insider threats, and zero-day exploits (Sharma & Bhushan, 2026; Waqdan *et al.*, 2025). Cross-layer analyses of resource-constrained IoT and Industrial IoT devices have identified firmware-level threats such as bootloader backdoors, hardcoded credentials, and unpatched CVE exploits, alongside network-level threats addressed through lightweight and graph-based intrusion detection (Altaibek *et al.*, 2025). Federated and cloud-integrated IoT deployments introduce additional threat vectors, including adversarial manipulation of AI-driven defences and privacy leakage through model updates (Almutairi & Sheldon, 2025; ElZemity & Arief, 2024). Systematic risk-assessment taxonomies further categorise these threats according

to the perspectives, methodologies, and frameworks used to analyse them, providing a structured basis for prioritising defensive investment (Waqdan *et al.*, 2025).

## 2.4. Intrusion Detection Systems

### 2.4.1. Signature-Based IDS

Signature-based IDS detect intrusions by matching observed traffic against a database of known attack signatures. While computationally efficient and highly accurate for known threats, such systems are structurally unable to detect novel or zero-day attacks whose signatures have not yet been catalogued (Hssayeni & Mahgoub, 2026).

### 2.4.2. Anomaly-Based IDS

Anomaly-based IDS instead learn a model of normal behaviour and flag deviations as potential intrusions, using either statistical, machine-learning, or deep-learning techniques (Sadhwani *et al.*, 2024; Verma & Ranga, 2020). This approach can, in principle, detect previously unseen attacks, but often at the cost of higher false-positive rates and a dependency on the availability of representative training data (Waqdan *et al.*, 2025).

### 2.4.3. Hybrid IDS

Hybrid IDS combine signature-based and anomaly-based components, or combine multiple learning paradigms, in an effort to capture the complementary strengths of each approach (Altunay & Albayrak, 2023; Gueriani *et al.*, 2024). Such systems can achieve strong detection performance across both known and unknown attack types, though at the cost of additional architectural and computational complexity (Al-Hejri *et al.*, 2026; S. Li *et al.*, 2026).

## 2.5. Machine Learning-Based IDS

Classical machine-learning classifiers, including Decision Trees, Random Forest, Support Vector Machines, k-Nearest Neighbours, Naïve Bayes, and gradient-boosting variants such as XGBoost, CatBoost, and LightGBM, have been widely applied to IoT intrusion detection (Sadhwani *et al.*, 2024; Verma & Ranga, 2020). Verma and Ranga (2020) conducted a comprehensive

comparative study of ML classifiers for detecting denial-of-service attacks against IoT devices, benchmarking classifiers on datasets such as CIDDS-001, UNSW-NB15, and NSL-KDD and evaluating inference latency directly on Raspberry Pi hardware, offering practical guidance for selecting classifiers under IoT hardware constraints. Sadhwani *et al.* (2024) compared Random Forest, Decision Tree, Extra Tree Classifier, SVM, and k-Nearest Neighbour models against a Deep Neural Network for binary and multi-class classification of Edge-IIoTset traffic, finding that the deep model achieved the strongest overall accuracy while the classical ML models remained competitive and considerably less resource intensive. The Kolmogorov–Arnold Network (KAN), a recent architecture using learnable activation functions, has also been hybridised with XGBoost to improve both accuracy and interpretability in IoT intrusion detection (Amouri *et al.*, 2024). Overall, classical ML models offer computational efficiency and interpretability advantages that make them attractive for constrained deployments, but they typically require substantial manual feature engineering and can struggle to capture complex temporal or sequential attack patterns (Sadhwani *et al.*, 2024; Verma & Ranga, 2020).

### 2.6. Deep Learning-Based IDS

Deep learning architectures, including Multilayer Perceptrons (MLP), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), Bidirectional LSTM (BiLSTM), and autoencoders, automatically learn hierarchical feature representations from network traffic, reducing reliance on manual feature engineering (Gueriani *et al.*, 2024; Sadhwani *et al.*, 2024). CNN architectures excel at extracting local spatial patterns from structured traffic features, while LSTM and GRU variants are well suited to modelling the temporal dependencies present in sequential network flows (Altunay & Albayrak, 2023; Gueriani *et al.*, 2024). Self-supervised and contrastive-learning variants of deep architectures have also been proposed to reduce dependence on labelled attack data: Z. Li and Yao (2024) combined self-supervised contrastive learning with self-knowledge distillation to produce a two-stage, lightweight IDS with reduced complexity and faster inference, while L. Li *et al.* (2024) used a hierarchical CNN-GRU model trained with a contrastive objective to improve detection of previously unseen attacks. Despite their strong representational capacity, deep

learning IDS generally require considerably more computation and memory than classical ML approaches, and their internal decision processes remain difficult to interpret without additional explainability tooling (Attique *et al.*, 2024).

### 2.7. Ensemble Learning for IoT Security

Ensemble learning techniques, including bagging, boosting, voting, and stacking, combine multiple base learners to improve robustness and accuracy relative to any single model (Lazzarini *et al.*, 2023; Wakili & Bakkali, 2025). Lazzarini *et al.* (2023) proposed a Deep Integrated Stacking model (DIS-IoT) that combines four deep-learning base learners through a fully connected fusion layer, achieving high accuracy with low false-positive rates on the ToN_IoT, CICIDS2017, and SWaT datasets. Wakili and Bakkali (2025) developed a fault-tolerant ensemble combining Balanced Random Forest, LightGBM, CatBoost, Extra Trees, and XGBoost with hybrid feature selection, achieving 99.16% accuracy on CICIoT2023 while explicitly analysing feature importance for explainability. Alani *et al.* (2024) combined BiLSTM and MLP classifiers with a decision-tree final-stage classifier, introducing a novel flow-based *history* feature to improve detection without excessive architectural complexity. Wang *et al.* (2026) proposed a three-model decision framework combining binary and multi-class neural networks with a k-nearest neighbour similarity model to explicitly identify novel, previously unseen attacks in addition to known attack types. Sanju (2023) combined an ensemble of recurrent neural networks (LSTM and GRU) with metaheuristic feature selection based on the Harris Hawk Optimisation algorithm. At the federated level, Chatterjee and Hanawal (2022) proposed a hybrid ensemble classifier (PHEC) later adapted to a federated learning setting and Jadhav *et al.* (2026) combined a residual autoencoder, attention capsule network, and stacked BiLSTM as base classifiers within an ensemble-based deep federated learning (E-DFL) framework, achieving 99.23% accuracy on Edge-IIoT data while preserving client privacy. Ensemble approaches have also been extended to adjacent domains such as the Internet-of-Energy (Moubayed, 2026) and smart grids (Aslam *et al.*, 2026). Across this literature, ensemble methods consistently improve detection robustness, but the aggregation of multiple learners inherently increases model size, training cost, and inference latency, which is a significant concern for constrained IoT deployment (Jadhav *et al.*,

2026; Lazzarini *et al.*, 2023).

### 2.8. Hybrid Deep Learning Models

Hybrid deep-learning architectures combine complementary neural components, most commonly CNN with LSTM, GRU, and BiLSTM, to jointly capture spatial and temporal patterns in network traffic (Agbor *et al.*, 2025; Altunay & Albayrak, 2023; Gueriani *et al.*, 2024). Gueriani *et al.* (2024) combined CNN and LSTM layers and evaluated the resulting model on the CICIoT2023 and CICIDS2017 datasets, reporting 98.42% accuracy with a low false-positive rate. Altunay and Albayrak (2023) compared a standalone CNN, a standalone LSTM, and a hybrid CNN+LSTM model on the UNSW-NB15 and X-IIoTID datasets, finding that the hybrid model consistently outperformed either component alone. Sadhwani *et al.* (2026) similarly combined BiLSTM and CNN components, evaluating the model on both GPU and CPU hardware to assess real-world IoT deployment feasibility. Agbor *et al.* (2025) proposed a CNN–BiLSTM–DNN hybrid evaluated on IoT-23 and Edge-IIoTset, applying pruning and quantisation to improve deployment efficiency on constrained hardware. Duraibi and Alashjaee (2026), S. Li *et al.* (2026), and Othman *et al.* (2026) each proposed variants of CNN–LSTM hybrids evaluated across multiple benchmark datasets (including BoT-IoT, CIC-IDS2017, ToN_IoT, and UNSW-NB15), consistently reporting strong accuracy alongside feature-engineering or fusion mechanisms intended to improve generalisation. Jouhari and Guizani (2024) specifically targeted resource-constrained deployment by proposing a lightweight CNN–BiLSTM hybrid designed to run onboard IoT hardware, reporting competitive accuracy on UNSW-NB15 with reduced model complexity. AL Ibaisi *et al.* (2026) introduced a bio-inspired amino-acid-style encoding combined with a two-stage hybrid feature-selection pipeline (Chi-Squared and CNN-Random Forest), achieving near-perfect accuracy with sub-2 ms inference latency on commodity hardware. Vision-transformer hybrids extend this line of work by fusing convolutional and attention-based components: C. Du *et al.* (2024) proposed MBConv-ViT, combining MobileNet-style convolution with a Vision Transformer, while Sana *et al.* (2024) combined tree-based ML, ensembles, LSTM, and Vision Transformers, reporting substantial performance gains from the transformer component. Al-Hejri *et al.* (2026) proposed an explainable hybrid CNN–Transformer model

integrated with federated learning and LIME-based interpretability, evaluated on IoT-23 and CIC IoT-DIAD 2024. Collectively, hybrid deep-learning models consistently outperform their individual constituent architectures, but the literature shows a clear pattern in which accuracy gains are accompanied by increased architectural and computational complexity, reinforcing the motivation for conditional-computation approaches such as sparse Mixture-of-Experts.

### 2.9. Graph Learning for IoT Security

Graph-based approaches represent network traffic as graphs of interacting hosts or flows, allowing Graph Neural Networks (GNN) and Graph Attention Networks to exploit relational structure that traditional flow-based features cannot easily capture (H. Nguyen & Kashef, 2023; R. Xu *et al.*, 2024). R. Xu *et al.* (2024) proposed a self-supervised GNN approach using graph contrastive learning to classify network flows by specific attack type without requiring extensive manual labelling, evaluated on NF-Bot-IoT and NF-CSE-CIC-IDS2018 variants. H. Nguyen and Kashef (2023) similarly proposed a traffic-aware self-supervised GNN (TS-IDS) that captures relationships between network entities using both node and edge features, evaluated on NF-ToN-IoT and NF-BoT-IoT. These approaches demonstrate that graph-based representations can improve detection of complex, multi-entity attack patterns and reduce reliance on labelled data, though they generally introduce additional computational overhead associated with graph construction and message-passing operations, which is a relevant consideration for constrained IoT deployment.

### 2.10. Transformer-Based IDS

Transformer architectures, built around self-attention mechanisms, have been increasingly applied to IoT intrusion detection because of their ability to model long-range dependencies within network traffic sequences (Hssayeni & Mahgoub, 2026; Sana *et al.*, 2024). Hssayeni and Mahgoub (2026) proposed a dual-stage Transformer IDS operating directly on raw packet payloads to detect zero-day attacks, evaluated on MQTT-IoT, IoT-23, and CIC-IoT-2022 under cross-validation settings designed specifically to test generalisation to unseen scenarios and networks. Almadhor *et al.* (2025) fine-tuned large transformer models, including BERT, Dis-

tilBERT, and RoBERTa, for anomaly detection on the RT_IoT2022 dataset, converting traffic into text representations to leverage pre-trained language understanding. J. Du (2026) combined a lightweight MobileNetV3 CNN with a Mini-Transformer for sequence modelling within an attention-augmented federated learning framework, evaluated on Edge-IIoTset and ToN-IoT. Adjewa *et al.* (2026) proposed SEED, a collaborative embedding-offloading mechanism in which a compact, fine-tuned BERT model generates traffic embeddings at the edge while a very small on-device neural network performs the final classification, substantially reducing the on-device model footprint while retaining high detection accuracy. Al-Nomasy *et al.* (2025) proposed DistillGuard, which uses a Transformer-based teacher model with hybrid multi-head and cross-attention layers, distilled into a lightweight student model via Selective Gradient-Based Knowledge Distillation, while also providing gradient-based explainability. Ailabouni *et al.* (2026) combined a Transformer Autoencoder with reinforcement-learning-tuned anomaly thresholding within a federated learning setup evaluated on BoT-IoT and custom IoT traffic. These studies collectively demonstrate that transformer-based IDS can achieve state-of-the-art detection accuracy, including strong zero-day and cross-dataset generalisation, but frequently require either substantial on-device computation or an edge-offloading architecture to remain deployable on constrained hardware (Adjewa *et al.*, 2026; Almadhor *et al.*, 2025).

### 2.11. Lightweight AI for Resource-Constrained IoT Devices

A distinct thread of research addresses the deployment gap between high-capacity deep learning models and the limited computational budgets of IoT hardware through lightweight AI and TinyML techniques, including model pruning, quantisation, and knowledge distillation (Agbor *et al.*, 2025; Al-Nomasy *et al.*, 2025; Jouhari & Guizani, 2024). Agbor *et al.* (2025) applied pruning and quantisation to a hybrid CNN–BiLSTM–DNN model to improve deployment efficiency while maintaining 99% accuracy on IoT-23 and Edge-IIoTset. Jouhari and Guizani (2024) designed a CNN–BiLSTM hybrid explicitly to satisfy the computation capability requirements of onboard IoT hardware. Z. Li and Yao (2024) used self-knowledge distillation to transfer learned representations from a larger teacher model to a smaller, faster student model, explicitly targeting reduced model complexity and increased inference speed. Adjewa *et al.* (2026)

achieved an approximately 90% reduction in model size for their edge-side BERT embedding model, with the final on-device classifier occupying only 137 KB, illustrating the scale of compression achievable through edge-offloading architectures. These approaches demonstrate multiple complementary strategies, compression, distillation, and architectural offloading, for reconciling deep-learning-level accuracy with IoT-level resource constraints, and they inform the lightweight expert design adopted in the present study.

### 2.12. Federated Learning for IoT Security

Federated learning (FL) enables multiple IoT clients to collaboratively train a shared intrusion-detection model without transmitting raw traffic data to a central server, thereby addressing privacy and bandwidth concerns associated with centralised IDS (Belarbi *et al.*, 2023; Chatterjee & Hanawal, 2022). Belarbi *et al.* (2023) evaluated an FL-based IDS under realistic, non-IID data-distribution conditions using the ToN-IoT dataset, finding that data heterogeneity considerably degrades performance relative to centralised training, though pre-training the global model improved F1-score by over 20%. T. D. Nguyen *et al.* (2026) proposed Fed-FeRe, integrating feature selection with a gated recurrent unit model and adaptive hyperparameter estimation, reducing communication cost relative to FedAvg while improving detection precision. Ilias *et al.* (2026) integrated a sparsely gated Mixture-of-Experts layer into a federated learning IDS, using noisy top-k gating to select a subset of experts and a label-based Dirichlet partitioning method to simulate non-IID client data, evaluated on the BoT-IoT dataset. Asiri (2026) proposed Causal-FL-ID, integrating local causal discovery at IoT clients with a globally aggregated causal graph on the server, enabling counterfactual reasoning about intrusions while preserving privacy. Sundaravadivel *et al.* (2026) combined federated deep reinforcement learning with a Transformer-based anomaly detector for agricultural IoT networks, validated across physical Raspberry Pi and virtualised edge testbeds. Fraihat *et al.* (2026) combined metaheuristic feature optimisation with a gated Transformer–LSTM model under an attention-weighted federated aggregation protocol (SwarmFed), evaluated on four IIoT datasets. Ailabouni *et al.* (2026) and J. Du (2026) likewise embedded transformer-based components within federated learning pipelines, while Jadhav *et al.* (2026) combined ensemble deep learning with feder-

ated aggregation. ElZemity and Arief (2024) conducted a systematic review of privacy threats specific to federated learning in IoT settings, cataloguing inference, poisoning, and eavesdropping attacks alongside defences such as differential privacy and secure multi-party computation. Collectively, this body of work shows that FL is a mature and actively developing paradigm for privacy-preserving IoT intrusion detection, and that Mixture-of-Experts has already begun to be integrated into FL-based IDS specifically to manage model complexity under non-IID conditions (Ilias *et al.*, 2026), although lightweight, deployment-aware expert design within this setting remains under-explored.

### 2.13. Explainable Artificial Intelligence

Explainable AI (XAI) techniques, including SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations), aim to make the predictions of otherwise opaque machine-learning and deep-learning models interpretable to human analysts (Attique *et al.*, 2024; Wakili & Bakkali, 2025). Attique *et al.* (2024) incorporated SHAP into a BiLSTM model with a self-adaptive attention mechanism to improve both data efficiency and transparency in IIoT attack detection. Wakili and Bakkali (2025) used explainability analysis to identify packet inter-arrival time, TCP protocol features, and traffic magnitude as the most discriminative indicators within their ensemble IDS, helping security teams interpret detection outcomes. Al-Nomasy *et al.* (2025) integrated gradient contribution heatmaps and layer-wise contribution analysis directly into a knowledge-distillation-based Transformer IDS. Asiri (2026) took a causal-reasoning approach to explainability, enabling counterfactual simulation of interventions rather than purely correlational feature attribution. Al-Hejri *et al.* (2026) applied LIME to a hybrid CNN–Transformer federated model to provide feature-level explanations supporting real-time analyst decision-making. Across these studies, explainability is shown to improve trust and support root-cause analysis, but it is still only sparingly integrated into advanced, high-accuracy IDS architectures, particularly those based on sparse or conditional computation.

## 2.14. Sparse Mixture-of-Experts

Sparse Mixture-of-Experts (MoE) architectures consist of a set of expert sub-networks together with a gating (routing) network that selects, for each input, only a small subset of experts to activate, thereby decoupling total model capacity from per-sample computational cost (Ilias *et al.*, 2026). Within the IoT intrusion-detection literature reviewed for this study, Ilias *et al.* (2026) represents the clearest example of MoE applied directly to this task: the authors integrated a sparsely gated MoE layer, comprising a router and a set of experts selected via noisy top-k gating, into a federated learning IDS explicitly to mitigate the model-complexity challenges introduced by non-IID client data, evaluating the approach on the BoT-IoT dataset and reporting competitive performance under heterogeneous federated conditions. Beyond this study, MoE-style conditional computation is echoed conceptually in edge-offloading architectures such as SEED, where a compact on-device network selectively processes embeddings generated by a larger, less frequently invoked model (Adjewa *et al.*, 2026), and in knowledge-distillation frameworks that similarly aim to concentrate computation where it is most needed (Al-Nomasy *et al.*, 2025; Z. Li & Yao, 2024). Foundation-model surveys covering the broader IoT domain likewise identify efficiency, alongside context-awareness, safety, and security, as one of the central shared objectives that techniques such as MoE are intended to address (Wei *et al.*, 2026). Taken together, the literature indicates that sparse MoE is, at present, applied to IoT intrusion detection in only a very limited number of studies, and almost exclusively within a federated learning context focused on managing data heterogeneity rather than on lightweight expert design, adaptive or resource-aware routing, or joint accuracy–efficiency evaluation, representing a clear research gap that this study addresses.

## 2.15. Public IoT Intrusion Detection Datasets

Recent IoT intrusion-detection research draws on a recurring set of public benchmark datasets. CICIoT2023 has been used to evaluate hybrid CNN–LSTM models (Gueriani *et al.*, 2024) and fault-tolerant ensemble frameworks (Wakili & Bakkali, 2025). Edge-IIoTset has been used to benchmark ML and DL classifiers (Sadhwani *et al.*, 2024), hybrid CNN–BiLSTM–DNN

models (Agbor *et al.*, 2025), attention-augmented federated transformer models (J. Du, 2026), and federated ensemble models (Jadhav *et al.*, 2026). IoT-23 has been used for hybrid CNN–BiLSTM–DNN evaluation (Agbor *et al.*, 2025), explainable hybrid CNN–Transformer federated models (Al-Hejri *et al.*, 2026), LLM-based threat-detection frameworks (Otoum *et al.*, 2025), and hybrid voting ensembles (Aslam *et al.*, 2026). ToN-IoT has been used to evaluate federated learning IDS under non-IID conditions (Belarbi *et al.*, 2023), stacking ensembles (Lazzarini *et al.*, 2023), and Transformer–LSTM federated frameworks (Fraihat *et al.*, 2026). BoT-IoT (and its NetFlow-derived variants, NF-BoT-IoT) has been used for federated MoE-based IDS (Ilias *et al.*, 2026), self-supervised GNN approaches (H. Nguyen & Kashef, 2023; R. Xu *et al.*, 2024), and federated Transformer-Autoencoder models (Ailabouni *et al.*, 2026). UNSW-NB15 has been used to benchmark hybrid CNN+LSTM models (Altunay & Albayrak, 2023), lightweight CNN–BiLSTM models (Jouhari & Guizani, 2024), and vision-transformer-augmented pipelines (Sana *et al.*, 2024). Additional datasets appearing in the reviewed literature include RT_IoT2022 (Almadhor *et al.*, 2025), MQTT-IoT and CIC-IoT-2022 (Hssayeni & Mahgoub, 2026), X-IIoTID (Altunay & Albayrak, 2023; Attique *et al.*, 2024), and CIC IoT-DIAD 2024 (Al-Hejri *et al.*, 2026). This diversity of datasets, while beneficial for cross-validating findings, also complicates direct comparison across studies, since feature schemas, attack taxonomies, and class distributions differ considerably between datasets.

### 2.16. Performance Evaluation Metrics

Classification performance in the reviewed literature is most commonly reported using accuracy, precision, recall, F1-score, and, in several recent studies, the Matthews Correlation Coefficient (MCC) and ROC-AUC, alongside detection rate and false-alarm or false-positive rate (Agbor *et al.*, 2025; Sana *et al.*, 2024; Wakili & Bakkali, 2025). Efficiency-oriented metrics are reported less consistently but appear in studies explicitly targeting resource-constrained deployment, including inference latency and throughput (Adjewa *et al.*, 2026; AL Ibaisi *et al.*, 2026), parameter counts and model size (Adjewa *et al.*, 2026; Agbor *et al.*, 2025), and CPU or memory utilisation during operation (Sadhwani *et al.*, 2026; Sundaravadivel *et al.*, 2026). Very few studies jointly report both detection-accuracy metrics and a full complement of efficiency

metrics (FLOPs, parameters, latency, and memory), which itself constitutes part of the evaluation gap addressed by this study.

## 2.17. Review of Recent Related Studies

### 2.17.1. Machine Learning-Based IDS

Recent ML-based studies continue to demonstrate that classical classifiers remain competitive on IoT traffic, particularly when combined with careful feature engineering or hardware-aware evaluation (Sadhwani *et al.*, 2024; Verma & Ranga, 2020). Hybridisation with more expressive function approximators, such as the KAN–XGBoost combination proposed by Amouri *et al.* (2024), further improves both accuracy and interpretability relative to standalone ML classifiers.

### 2.17.2. Deep Learning-Based IDS

Deep-learning studies published between 2023 and 2026 emphasise self-supervised and contrastive-learning techniques to reduce labelling requirements (L. Li *et al.*, 2024; Z. Li & Yao, 2024), alongside continued refinement of CNN, LSTM, and BiLSTM-based architectures for both binary and multi-class attack classification (Gueriani *et al.*, 2024).

### 2.17.3. Ensemble Learning Models

Ensemble studies in this period span stacking (Lazzarini *et al.*, 2023), voting (Alani *et al.*, 2024; Aslam *et al.*, 2026), and federated ensemble designs (Chatterjee & Hanawal, 2022; Jadhav *et al.*, 2026), consistently reporting accuracy improvements over single-model baselines, generally at the cost of increased architectural complexity (Wakili & Bakkali, 2025).

### 2.17.4. Hybrid Deep Learning Models

Hybrid CNN–LSTM/BiLSTM architectures represent the most densely populated theme in the recent literature, with numerous independent studies reporting accuracies exceeding 97–99% across datasets such as BoT-IoT, UNSW-NB15, ToN-IoT, and CICIoT2023 (Agbor *et al.*, 2025; AL Ibaisi *et al.*, 2026; Altunay & Albayrak, 2023; Debnath & Das, 2026; Duraibi & Alashjaee, 2026; Ikhlas *et al.*, 2026; Jouhari & Guizani, 2024; S. Li *et al.*, 2026; Othman *et al.*, 2026;

Sadhwani *et al.*, 2026). Vision-transformer hybrids extend this theme with additional attention-based components (C. Du *et al.*, 2024; Sana *et al.*, 2024).

### 2.17.5. Federated Learning-Based IDS

Federated approaches published in this window increasingly integrate advanced components, feature reduction (T. D. Nguyen *et al.*, 2026), causal reasoning (Asiri, 2026), reinforcement-learning-tuned thresholds (Ailabouni *et al.*, 2026), deep reinforcement learning with transformers (Sundaravadivel *et al.*, 2026), and metaheuristic-optimised gated fusion (Fraihat *et al.*, 2026), reflecting a maturing but still rapidly evolving research area.

### 2.17.6. Graph Learning-Based IDS

Graph-based studies in this period emphasise self-supervised and contrastive objectives to reduce dependence on labelled data while modelling relational structure between network flows (H. Nguyen & Kashef, 2023; R. Xu *et al.*, 2024).

### 2.17.7. Transformer-Based IDS

Transformer-based studies range from raw-payload dual-stage transformers targeting zero-day detection (Hssayeni & Mahgoub, 2026) to fine-tuned large pre-trained language models (Almadhor *et al.*, 2025; Liu *et al.*, 2025; Ma *et al.*, 2025), lightweight mini-transformer hybrids (J. Du, 2026), and edge-offloaded transformer embeddings (Adjewa *et al.*, 2026).

### 2.17.8. Lightweight AI and TinyML-Based IDS

Studies explicitly targeting lightweight deployment include pruning and quantisation of hybrid models (Agbor *et al.*, 2025), purpose-built lightweight CNN–BiLSTM hybrids (Jouhari & Guizani, 2024), self-knowledge distillation (Z. Li & Yao, 2024), and edge-embedding offloading with drastic model-size reduction (Adjewa *et al.*, 2026).

### 2.17.9. Explainable AI-Based IDS

Explainability-focused studies integrate SHAP-based feature attribution (Attique *et al.*, 2024; Wakili & Bakkali, 2025), gradient-based layer attribution within knowledge distillation (Al-Nomasy *et al.*, 2025), causal explanation (Asiri, 2026), and LIME-based local explanation within federated hybrid models (Al-Hejri *et al.*, 2026).

### 2.17.10. Sparse Mixture-of-Experts and Foundation Models

As discussed in Section 2.14, MoE-based IDS research within this window is represented chiefly by Ilias *et al.* (2026), situated within a federated learning context, while broader IoT foundation-model surveys (Wei *et al.*, 2026) and adjacent LLM-for-cybersecurity surveys (Alqahtani & Ahuja, 2026; Alqahtani & Kumar, 2026; Chen *et al.*, 2024; Das *et al.*, 2025; Gökstorp *et al.*, 2026; Nageshwaran & Ezekiel, 2026; Vyas *et al.*, 2025; H. Xu *et al.*, 2025) provide relevant context on efficiency, safety, and governance considerations for large and sparse models applied to security tasks, alongside LLM-based threat-detection frameworks applied directly to IoT (Al-Hammouri *et al.*, 2025; Otoum *et al.*, 2025).

### 2.17.11. Comparative Analysis of Existing Studies

Table 2.1 summarises representative studies across the themes reviewed above, highlighting the model/technique employed, the dataset(s) used, reported performance, and the extent to which computational cost was evaluated.

Table 2.1: Comparative summary of recent studies

| **Author(s)** | **Year** | **Model/Technique** | **Dataset** | **Acc./F1** | **Computational Cost Reported** |
|---|---|---|---|---|---|
| Verma and Ranga (2020) | 2020 | ML classifier benchmark | CIDDS-001, UNSW-NB15, NSL-KDD | Varies by classifier | Yes (Raspberry Pi latency) |
| Gueriani *et al.* (2024) | 2024 | CNN–LSTM hybrid | CICIoT2023, CICIDS2017 | 98.42% acc. | Loss/FPR only |

| | | | | | |
|---|---|---|---|---|---|
| Z. Li and Yao (2024) | 2024 | Contrastive learning + self-distillation | KDD99, NSL-KDD, UNSW-NB15, CICIDS2017, BoT-IoT | State-of-the-art | Speed/complexity discussed |
| Lazzarini *et al.* (2023) | 2023 | Stacking ensemble (DIS-IoT) | ToN_IoT, CICIDS2017, SWaT | High acc., low FPR | Not primary focus |
| Ilias *et al.* (2026) | 2026 | Federated sparse MoE | BoT-IoT | Competitive across settings | Model complexity discussed |
| Hssayeni and Mahgoub (2026) | 2026 | Dual-stage Transformer | MQTT-IoT, IoT-23, CIC-IoT-2022 | +44% / +16% F1 over baseline | Not primary focus |
| Adjewa *et al.* (2026) | 2026 | Edge-offloaded BERT + tiny NN (SEED) | Custom edge traffic | 99.9% acc. | Yes (model size, latency) |
| Al-Nomasy *et al.* (2025) | 2025 | Transformer teacher–student distillation (DistillGuard) | IoT traffic | Superior to baselines | Yes (distillation cost) |
| Agbor *et al.* (2025) | 2025 | CNN–BiLSTM–DNN | IoT-23, Edge-IIoTset | 99% acc. | Yes (pruning, quantisation) |
| Wakili and Bakkali (2025) | 2025 | Fault-tolerant ensemble | CICIoT2023 | 99.16% acc. | Feature-level, not compute |

## 2.18. Research Gaps

Synthesising the literature reviewed above, several research gaps can be identified:

(a) Most IDS models prioritise detection accuracy over computational efficiency, limiting their deployment on resource-constrained IoT devices (Adjewa *et al.*, 2026; Almadhor *et al.*, 2025).

(b) Transformer-based IDS often incur high computational and memory overhead, even when they achieve strong detection accuracy (C. Du *et al.*, 2024; Hssayeni & Mahgoub, 2026; Sana *et al.*, 2024).

(c) Ensemble and hybrid models consistently improve detection performance but typically increase model complexity and inference latency rather than reducing it (Agbor *et al.*, 2025; Lazzarini *et al.*, 2023).

(d) Sparse Mixture-of-Experts has seen very limited application in IoT intrusion detection, with the reviewed literature identifying essentially a single directly relevant study, situated specifically within a federated learning context (Ilias *et al.*, 2026).

(e) Few studies jointly address lightweight expert design, adaptive or resource-aware routing, and intrusion detection performance within a single evaluation framework.

(f) Explainability is not commonly integrated into advanced, high-accuracy IDS architectures, and where it is integrated, it is rarely combined with sparse or conditional-computation models (Al-Hejri *et al.*, 2026; Al-Nomasy *et al.*, 2025; Attique *et al.*, 2024).

(g) There is a lack of standardised evaluation that jointly balances security performance with resource consumption metrics such as latency and memory (Adjewa *et al.*, 2026; Sundaravadivel *et al.*, 2026).

These gaps collectively motivate the proposed *Lightweight Sparse Mixture-of-Experts Intrusion Detection Framework*, which aims to balance high detection performance, computational efficiency, scalability, and practical deployability on resource-constrained IoT devices, extending the single existing federated MoE study in this space (Ilias *et al.*, 2026) towards lightweight, heterogeneous, and deployment-aware expert design.

# CHAPTER THREE

# MATERIALS AND METHODS

This chapter presents the materials and methods that supports the proposed lightweight sparse Mixture of Experts (MoE) framework for intrusion detection in resource constrained Internet of Things (IoT) networks. The chapter opens by presenting the materials used throughout this research, and situating the study within an empirical, design science research paradigm and by describing the end to end experimental workflow. It then documents the computational environment, the deterministic execution regime that guarantees repeatability, and the two benchmark datasets that anchor the evaluation, namely IoT-23 and Edge-IIoTset. Thereafter it develops the preprocessing pipeline in full, including the removal of uninformative, identifying and target leaking attributes, per class outlier treatment by Tukey winsorisation, feature scaling, class rebalancing by minority only synthetic oversampling, and the construction of pseudo sequential tensors for the convolutional and recurrent baselines. The core of the chapter formalises the proposed architecture: a shared feature embedding, a pool of eight architecturally heterogeneous experts, a noisy top-$k$ router regularised by a load balancing auxiliary objective, and a fusion head that recombines the sparsely weighted expert responses. Every design decision is justified with reference both to the computational constraints of edge deployment and to the empirical behaviour observed during development. The chapter also specifies the seven comparator models, the optimisation and validation protocol, the four level explainability framework, the robustness protocol based on withheld attack families, the ablation designs, the evaluation metrics and the computational complexity accounting. The chapter closes by consolidating the reproducibility measures and by transitioning into the empirical results reported in Chapter 4.1.

## 3.1. Materials

### 3.1.1. Hardware and Software Resources

In this research, all experiments are performed on the Dell Precision 7740, equipped with Xeon processor, NVIDIA RTX 4000 (Quadro) Graphics Processing Unit (GPU) to accelerate the training, validation, and testing of the proposed intrusion detection models. In addition to GPU, the system used for this research is also equipped with 64 GB of memory with a SWAP of 256 GB to facilitate large dataset processing facilitates efficient implementation. The proposed framework is implemented in Python using the Jupyter Notebook interface available installed as an extension in Visual Studio Code (VS Code). Python (v3.11.5) is selected due to its rich collection of scientific computing, machine learning, and deep learning libraries, which support efficient data preprocessing, feature engineering, model implementation, performance assessment, and result visualisation.

The principal software tools utilised in this study are summarised below:

**NumPy**

is employed for high-performance numerical computing and efficient manipulation of multidimensional arrays during data preparation and model implementation.

**Pandas**

is used for loading, cleaning, integrating, transforming, and managing the network traffic datasets and their associated metadata throughout the data preprocessing stage.

**Matplotlib**

Is utilised to visualise dataset characteristics, model performance, learning curves, computational efficiency metrics, and explainability results.

**Scikit-learn**

provides utilities for data preprocessing, feature scaling, dataset partitioning, performance evaluation, and the implementation of conventional machine learning models that serve as baseline methods for comparison.

**TensorFlow with the Keras API**

serves as the primary deep learning framework for designing, training, validating, and optimising the proposed Sparse Mixture-of-Experts intrusion detection framework and the baseline deep learning models used for comparative evaluation.

**Jupyter Notebook**

provides an interactive programming environment for implementing algorithms, preprocessing datasets, training and evaluating models, and documenting the experimental workflow.

## 3.2. Methods

This section focuses on providing the details of the methodology adopted to design, implement, and evaluate the proposed lightweight sparse mixture-of-experts (MoE) intrusion detection framework starting with the research design (Section 3.3). Consistent with the research gaps identified in Chapter 2, the methodology is deliberately structured so that the novelty of the study rests not merely on the use of a mixture-of-experts architecture, since this has already been explored in a federated learning context (Ilias *et al.*, 2026), but on the combination of heterogeneous, lightweight experts; resource-aware conditional routing, and joint accuracy-efficiency evaluation. The section covers the research design, the datasets and preprocessing pipeline, the baseline models against which the proposed framework is compared, the architecture of the proposed framework itself, the training procedure, the experimental design mapped explicitly to the research questions stated in Chapter 1, the evaluation metrics, the planned ablation studies, the explainability procedure, and the tools and implementation environment.

## 3.3. Research Design

The study adopts a quantitative, experimental research design executed within the design science research paradigm, in which an artefact, here a novel intrusion detection model, is constructed, instantiated and then rigorously evaluated against explicit performance and efficiency criteria. The design science orientation is appropriate because the central research questions are simultaneously constructive and evaluative: they ask not only whether an accurate detector can be built,

but whether that detector can be made lightweight enough to run within the memory, latency and energy envelope of constrained IoT hardware while remaining interpretable and robust. This dual objective cannot be settled by observation alone; it requires the deliberate construction of an artefact whose internal structure is instrumented so that both its predictive behaviour and its computational cost can be measured directly.

Epistemologically, the work is positivist in that it treats detection performance, inference latency, parameter count and floating point operation (FLOP) counts as objective, measurable quantities whose values are established through controlled experimentation rather than through interpretation. Every claim advanced in Chapter Four is therefore expressed as a measured quantity derived from a fixed, seeded pipeline and is accompanied by the experimental configuration that produced it. The experimental unit is a single network flow record; the treatments are the competing models; and the response variables are the detection metrics and efficiency metrics defined in Section 3.13. Because the same preprocessing pipeline, the same stratified data partitions and the same random seed are shared across all models within a given dataset and task, differences in the response variables can be attributed to differences between the models rather than to incidental variation in data handling, which is the design principle that gives the comparative results their internal validity.

The design is further strengthened by a within study replication structure. Each experiment is executed in two classification regimes, binary detection (benign versus attack) and multiclass attack categorisation, and is repeated across two datasets that differ in their capture methodology, their feature vocabulary and their attack taxonomies. This factorial structure, two datasets crossed with two tasks, yields four primary experimental configurations and allows the study to distinguish findings that are idiosyncratic to a single corpus from findings that generalise across heterogeneous IoT traffic. The remainder of this chapter operationalises this design, beginning with the experimental workflow that binds the individual components together.

### 3.4. Experimental Workflow

This section focuses on providing the details of the methodology adopted to design, implement, and evaluate the proposed lightweight sparse mixture-of-experts (MoE) intrusion detection

framework starting with the research design (Section 3.3). Consistent with the research gaps identified in Chapter 2, the methodology is deliberately structured so that the novelty of the study rests not merely on the use of a mixture-of-experts architecture, since this has already been explored in a federated learning context (Ilias *et al.*, 2026), but on the combination of heterogeneous, lightweight experts; resource-aware conditional routing, and joint accuracy-efficiency evaluation. The section covers the research design, the datasets and preprocessing pipeline, the baseline models against which the proposed framework is compared, the architecture of the proposed framework itself, the training procedure, the experimental design mapped explicitly to the research questions stated in Chapter 1, the evaluation metrics, the planned ablation studies, the explainability procedure, and the tools and implementation environment.

The experimental workflow proceeds through a fixed sequence of stages that is identical across datasets and tasks, with only dataset specific parameters varying between runs. The workflow begins with raw data ingestion, in which the flow export of each dataset is loaded and concatenated. It continues with a cleaning stage that removes uninformative, identifying and leaking attributes; a numeric coercion and encoding stage that renders every retained attribute machine readable; an outlier treatment stage that winsorizes extreme continuous values within each class; an exploratory stage that records the class distribution; and a partitioning stage that produces stratified training, validation and test splits. Feature scaling is fitted on the training split alone and applied to the validation and test splits to prevent information leakage across the partition boundary. Minority class synthetic oversampling is then applied to the training split only, again to avoid contaminating the evaluation partitions. The processed tensors feed a model development stage in which the proposed sparse MoE and all seven baselines are trained under a shared optimisation protocol, followed by an evaluation stage that computes the full battery of detection and efficiency metrics on the held out test set. The workflow terminates with an interpretation stage comprising routing analysis, global and local explainability, robustness testing and ablation. Figure 3.1 depicts this pipeline schematically.

A defining property of this workflow is that all data dependent estimators, namely the feature scaler, the synthetic oversampler and every model, observe only the training partition during fitting. This discipline is not merely conventional hygiene; in the intrusion detection

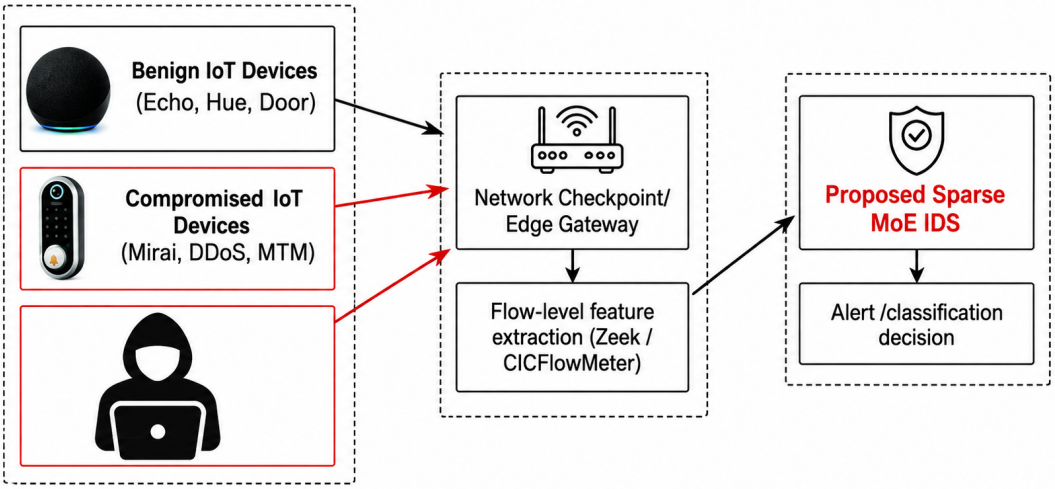



Figure 3.1: Experimental workflow of the proposed framework

setting it is essential to the validity of the reported generalisation, because IoT traffic corpora are frequently captured on a single testbed over a small number of scheduled sessions, and any estimator that is permitted to observe the test partition during fitting will silently absorb capture specific structure and report optimistically biased performance. The workflow therefore encodes leakage prevention as a structural invariant rather than as an afterthought, and the same principle motivates the leakage aware feature selection described in Section 3.7.1. Having established the sequence of operations, the next section specifies the environment in which they were executed.

### 3.5. Computational Environment and Reproducibility

All experiments were implemented in Python and executed within a Jupyter based scientific computing environment. Deep learning components were built on TensorFlow with its bundled Keras application programming interface, the classical machine learning baselines on scikit-learn, gradient boosting on XGBoost, minority oversampling on the imbalanced-learn library, and the post hoc explainability analyses on the SHAP and LIME libraries. Numerical computation and tabular manipulation were handled by NumPy and pandas respectively, and visualisation by Matplotlib and seaborn. Table 3.1 enumerates the principal software dependencies, and Table 3.2 summarises the hardware configuration on which the reported results were produced.

Table 3.1: Software environment and key dependencies

| **Component** | **Role in the pipeline** |
|---|---|
| Python 3 | Host programming language for the entire pipeline |
| TensorFlow / Keras | Construction and training of the sparse MoE and the deep, hybrid and transformer baselines |
| scikit-learn | Random Forest and Support Vector Machine baselines, preprocessing utilities, metrics and permutation importance |
| XGBoost | Gradient boosted decision tree baseline |
| imbalanced-learn (SMOTE) | Minority only synthetic oversampling of the training partition |
| SHAP | KernelSHAP global and local attributions for the proposed model; TreeSHAP reference for Random Forest |
| LIME | Local, per instance surrogate explanations |
| NumPy, pandas | Numerical arrays and tabular data manipulation |
| Matplotlib, seaborn | Generation of all figures and diagnostic plots |

Table 3.2: Hardware and execution configuration

| **Item** | **Specification** |
|---|---|
| Accelerator | Graphics processing unit used for neural network training and inference |
| Host memory | Sufficient to hold the capped training corpora in memory during oversampling and scaling |
| Operating system | Linux based scientific computing host |
| Numerical precision | Single precision (float32) for all neural network weights and activations |
| Determinism | Operation level determinism enabled across CPU and GPU kernels |

Reproducibility was enforced through a single global random seed, fixed at the value 42, that was propagated to the Python hash seed, the NumPy generator, the Python `random` module and the TensorFlow and Keras random state. In addition, operation level determinism was enabled in TensorFlow, deterministic cuDNN kernels were requested, the oneDNN opportunistic optimisations that can perturb numerical results were disabled, and the Keras utility that seeds all internal generators was invoked. The combination of a fixed seed with operation level determinism means that the entire pipeline, from the stratified partitioning through the synthetic oversampling to the stochastic gradient updates and the noisy gate perturbations of the router, is reproducible run to run on the same hardware. This is a stronger guarantee than seed fixing alone, because many accelerated kernels are non associative under parallel reduction and will otherwise yield run to run variation even when seeded. Every random dependent operation in the study, including the subsampling of instances for KernelSHAP, the selection of samples for the per prediction routing inspector and the subsampling used to keep Support Vector Machine training tractable, draws from a seeded generator so that the identical instances are selected on every execution. Having fixed the environment, the study next specifies the datasets that populate the pipeline.

### 3.6. Benchmark Datasets

The evaluation is anchored on two publicly documented intrusion detection corpora that were selected to span complementary regions of the IoT and cyber physical threat landscape. IoT-23 provides Zeek connection log flows captured from consumer IoT devices infected with real malware families; and Edge-IIoTset provides a packet derived corpus of industrial IoT traffic spanning a broad taxonomy of attacks. Using two corpora with different capture tools, different native feature vocabularies and different attack taxonomies is a deliberate methodological choice: a framework that performs consistently across both is far more credible as a general purpose IoT intrusion detector than one validated on a single corpus, and the diversity of the corpora also exposes dataset specific artefacts that a single dataset study would leave hidden. Table 3.3 summarises the salient properties of the two corpora as they are used in this study.

Table 3.3: Benchmark datasets and classification tasks

| Dataset | Capture and feature vocabulary | Binary | Multiclass |
|---|---|---|---|
| IoT-23 | Zeek connection logs from malware infected consumer IoT devices | Normal vs. Attack | 5 classes |
| Edge-IIoTset | Packet derived industrial IoT flows over a broad attack taxonomy | Normal vs. Attack | 12 classes |

### 3.6.1. IoT-23

IoT-23 is delivered as Zeek connection logs whose native schema records, for each bidirectional flow, timing fields, source and destination host and port identifiers, transport protocol, application service, connection state, history flags and a family of byte and packet counters for the originator and responder directions. In this study the host and session identifiers, namely the timestamp, the originator and responder host addresses and the originator port, are removed because they encode which device generated a flow and when it was captured rather than how the traffic behaved, and retaining them would allow a model to memorise the fixed addressing of the capture testbed. The responder port is not discarded outright but is bucketed into a categorical port group, since the service implied by a destination port carries genuine behavioural signal whereas the raw port number is a high cardinality identifier. The connection state field is retained with missing values imputed to an explicit unknown category, the history string is dropped, and the connection state and port bucket are label encoded while the transport protocol and service are one hot encoded with the first level dropped to avoid collinearity. The retained numerical attributes are the flow duration, the originator and responder byte counts, the originator and responder IP layer byte counts, the originator and responder packet counts and the missed byte count. For tractability the corpus is capped so that no class exceeds one hundred thousand records, and the label vocabulary is consolidated: the several command and control variants are merged into a single command and control class, the horizontal port scan labels are mapped to a single port scan class, the file download traffic is excluded, and the remaining labels are standardised. The binary task contrasts normal traffic against all attack traffic, while the multiclass task distinguishes Benign, command and control, distributed denial of service, the Okiru botnet family and port scanning.

### 3.6.2. Edge-IIoTset

Edge-IIoTset is a packet derived industrial IoT corpus with a broad attack taxonomy. Its preprocessing performs targeted feature engineering rather than wholesale deletion: application layer payload text is mined for structured query language injection keywords, which are condensed into compact indicator features before the raw high cardinality payload text is discarded; destination ports are binned into service groups rather than deleted, so that the protocol level signal they carry is preserved without their identifier level noise; and pure identifier and noise attributes, namely the frame timestamp, the source and destination host names and the address resolution protocol IPv4 fields, are dropped outright. The largest classes are downsampled to a fixed ceiling to keep training tractable while preserving the natural minority structure of the rarer attack families. The binary task contrasts normal traffic against attack traffic, and the multiclass task distinguishes twelve categories, namely Backdoor, HTTP distributed denial of service, ICMP distributed denial of service, TCP distributed denial of service, UDP distributed denial of service, Fingerprinting, man in the middle, Normal, Password attacks, Ransomware, Reconnaissance and Web attack. Having specified the two corpora, the chapter now develops the common preprocessing pipeline in full.

## 3.7. Data Preprocessing Pipeline

The preprocessing pipeline transforms the heterogeneous raw flow exports into clean, numeric, appropriately scaled and class balanced tensors while preventing information leakage across the partition boundary. The pipeline is organised into leakage aware feature selection, numeric coercion and categorical encoding, per class outlier winsorization, class distribution analysis, stratified partitioning, training split feature scaling and minority only synthetic oversampling. Each stage is now described together with the scientific rationale for its inclusion.

### 3.7.1. Removal of Uninformative, Identifying and Leaking Attributes

The first and most consequential preprocessing stage removes attributes that either carry no information, identify who generated a flow and where or when it was captured, or directly leak the prediction target. This stage is decisive for the external validity of the entire study, because

network intrusion corpora are typically captured on a single testbed over a small set of scheduled sessions, and any attribute that encodes the fixed addressing, the capture ordering or the attack scheduling of that testbed provides a shortcut that a sufficiently expressive model will exploit in place of learning transferable attack behaviour. Four families of attributes are therefore removed, in a fixed order, before any feature engineering.

The first family comprises attributes with excessive missingness, removed when more than sixty percent of their values are absent, with the missing proportion computed dynamically from the loaded data and reported per attribute rather than assumed. The second family comprises constant attributes, identified by a unique value count of one, which by construction cannot discriminate between classes. The third family comprises host and session identifiers, namely source and destination addresses and ports, per packet identifiers, wall clock timestamps that encode the scheduled attack windows, capture position markers, idle time markers and constant deployment fields such as VLAN tags. A final step removes exact duplicate feature columns that arise when several flow statistics collapse to identical values because each record aggregates a single transaction. This leakage aware selection is applied consistently across the two corpora with dataset specific attribute names, and it embodies the same structural leakage prevention principle that governs the partitioning and scaling stages.

### 3.7.2. Numeric Coercion and Categorical Encoding

After feature selection, every retained attribute is rendered numeric. Continuous attributes are coerced to floating point, categorical attributes with an inherent ordering or a small vocabulary are label encoded, and nominal attributes such as the transport protocol and the application service are one hot encoded with the first level dropped to avoid the dummy variable collinearity that would otherwise destabilise the linear components of the models. A safety net pass verifies that the feature matrix is fully numeric before any statistical treatment is applied, since the subsequent outlier and scaling operations assume numeric input. The target column is constructed last and in a task aware manner, so that the remainder of the pipeline is agnostic to whether the run is binary or multiclass, and the raw label source columns are dropped immediately after the target is derived so that neither can leak into the feature matrix.

### 3.7.3. Per Class Outlier Treatment by Winsorisation

Network flow statistics are heavy tailed: a small fraction of flows carry byte and packet counts orders of magnitude larger than the median, and these extreme values can dominate the gradient signal and distort feature scaling. To temper this without discarding data, the pipeline applies Tukey winsorisation within each class. For each continuous attribute, defined as an attribute with more than twenty unique values, the first and third quartiles $Q_1$ and $Q_2$ are computed within each class, the interquartile range is formed as $\mathrm{IQR} = Q_3 - Q_1$, and the fence is set at

$$[\, Q_1 - k \cdot \mathrm{IQR},\ Q_3 + k \cdot \mathrm{IQR}\,], \qquad k = 1.5. \tag{3.1}$$

Values falling outside the fence are clipped to the nearest fence boundary rather than removed, so that every row and every class is preserved and the class distribution is left intact. Computing the fence within each class rather than globally is important, because the benign and attack populations have genuinely different central tendencies for many byte and packet statistics, and a global fence would misclassify the legitimate bulk of one class as outliers of the other. The per attribute outlier proportions are recorded for audit, and for the IoT-23 originator IP byte attribute, for example, roughly one flow in six lies beyond the fence, which quantifies the heavy tailed nature of the raw counts and motivates the winsorisation. The before and after distribution of a representative capped attribute is exported for each dataset for verification.

### 3.7.4. Stratified Partitioning and Training Split Feature Scaling

The processed corpus is partitioned into training, validation and test subsets using stratified sampling that preserves the class proportions in every subset. Thirty percent of the data is first held out, and that holdout is split evenly into validation and test subsets, yielding an approximate seventy, fifteen, fifteen percent partition. Stratification is essential given the class imbalance, because a naive random split would risk leaving the rarest attack families entirely absent from the validation or test subsets and would render their per class metrics undefined or unstable. Feature scaling by standardisation, subtracting the mean and dividing by the standard deviation of each attribute, is then fitted on the training subset alone and applied unchanged to the validation and

test subsets. Fitting the scaler on the training subset only is a strict leakage prevention measure: allowing the scaler to observe the test subset would let global distributional information from the evaluation data influence the model inputs and would inflate the reported generalisation.

### 3.7.5. Minority Only Synthetic Oversampling

Class imbalance is addressed on the training subset alone by the Synthetic Minority Oversampling Technique (SMOTE), which synthesises new minority instances by interpolating between existing minority instances and their nearest neighbours in feature space. The pipeline computes the median class frequency of the training subset and identifies as minority any class whose frequency falls below one tenth of that median; for each such class it synthesises additional instances up to a target of thirty percent of the median, using five nearest neighbours. Two design choices are important. First, oversampling is applied only to genuine minorities and only up to a modest target rather than to full balance, which augments the learning signal for the rarest families without swamping the training subset with synthetic points that could distort the decision boundary. Second, oversampling is applied strictly after partitioning and strictly to the training subset, so that no synthetic instance derived from a test flow ever appears in training and the test subset remains a faithful sample of the natural, imbalanced traffic distribution. This is the counterpart, on the label side, of the training only scaling discipline on the feature side. The processed training tensors that emerge from this stage feed both the proposed model and the baselines.

### 3.7.6. Pseudo Sequential Tensor Construction

The convolutional, recurrent, hybrid and transformer components operate on sequential input, whereas the flow records are tabular. To bridge this, each scaled feature vector of length $F$ is reshaped into a pseudo sequence of shape $(F, 1)$, treating the ordered features as positions along a one dimensional axis. This reshaping is applied identically to the proposed model and to the sequence oriented baselines, so that all sequence consuming models observe the same input representation. It should be emphasised that this axis is a feature axis and not a temporal axis; the reshaping enables convolutional and attention operators to act over local groups of features

but does not manufacture genuine temporal structure. This distinction is drawn upon in Chapter Four when interpreting why the sequence oriented baselines do not automatically outperform the classical tabular learners on these shuffled flow features. With the data representation established, the chapter now formalises the proposed architecture.

## 3.8. The Proposed Lightweight Sparse Mixture of Experts Framework

The proposed framework is a sparse Mixture of Experts classifier designed expressly for the resource envelope of constrained IoT hardware. Its central idea is conditional computation: rather than evaluating one large monolithic network for every flow, the model maintains a pool of small, architecturally diverse experts and, for each flow, activates only a small subset of them under the control of a learned router. Because only a fraction of the experts run per sample, the model can hold a large and diverse total capacity while incurring a per sample compute cost comparable to that of a single small network. This section formalises the shared embedding, the heterogeneous expert pool, the noisy top-$k$ router, the load balancing objective and the fusion head, and it explains the lightweight configuration that adapts the architecture to the edge. The overall architecture is depicted in Figure 3.2.

### 3.8.1. Problem Formulation and Notation

Let a preprocessed flow be represented by a feature vector $\mathbf{x} \in \mathbb{R}^F$, reshaped to the pseudo sequence $\mathbf{X} \in \mathbb{R}^{F\times 1}$ as described in Section 3.7.6, and let $y \in \{0, 1, \ldots, C-1\}$ be its class label, where $C = 2$ for the binary task and $C > 2$ for the multiclass task. The framework learns a mapping $f_\theta : \mathbb{R}^{F\times 1} \to \Delta^{C-1}$ from a flow to a probability distribution over the $C$ classes, where $\Delta^{C-1}$ is the probability simplex, and the predicted label is $\hat{y} = \arg\max_c f_\theta(\mathbf{X})_c$. The model comprises a shared embedding $g_\phi$, a set of $N$ experts $\{e_j\}_{j=1}^{N}$ with $N = 8$, a router $r_\psi$, a fusion head $h_\omega$ and a classifier, whose parameters are collected in $\theta$.

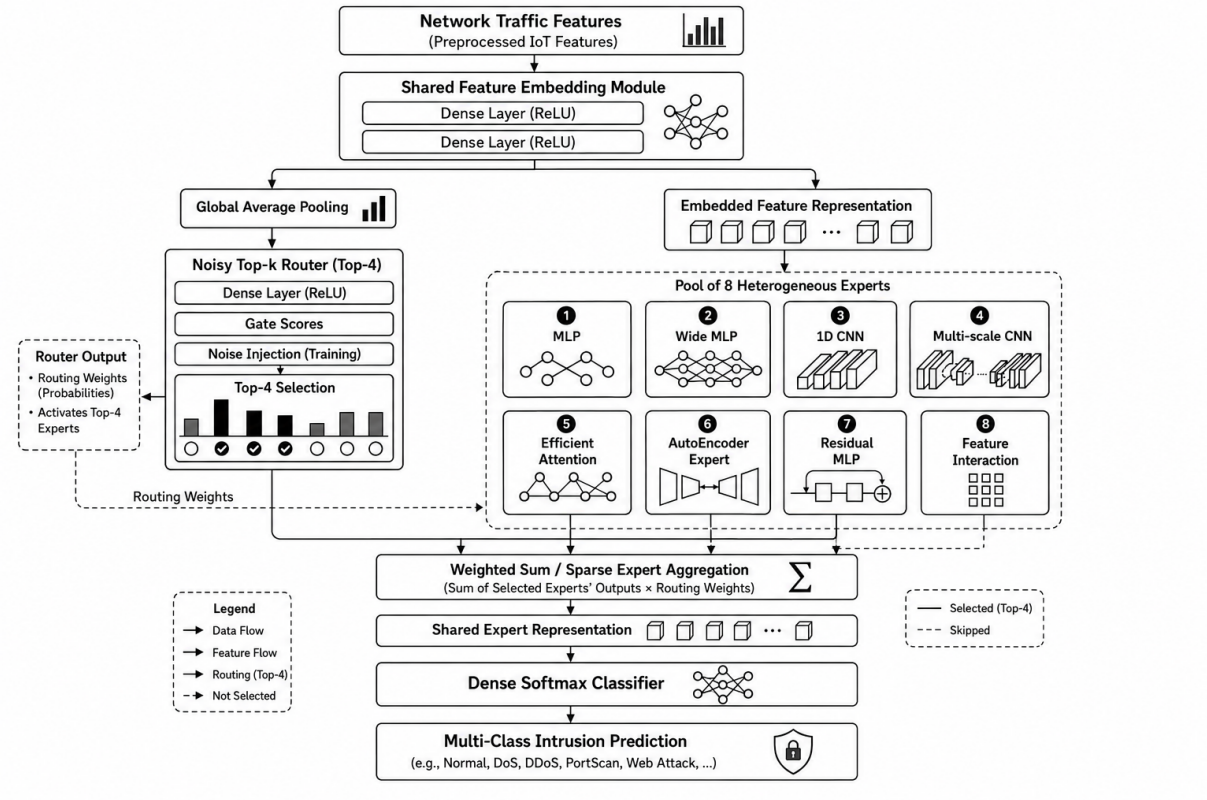



Figure 3.2: Architecture of the proposed lightweight sparse MoE detector

### 3.8.2. Shared Feature Embedding

Every flow first passes through a shared two layer feature embedding that projects each feature position into a $d$ dimensional space, with $d = 16$ in the lightweight configuration:

$$\mathbf{Z} = g_\phi(\mathbf{X}) = \sigma\big(\mathbf{W}_2\,\sigma(\mathbf{W}_1\mathbf{X})\big) \in \mathbb{R}^{F\times d}, \tag{3.2}$$

where $\sigma$ denotes the rectified linear activation. The embedding is deliberately two layers deep rather than a single linear projection, so that the representation presented to the experts is already partially nonlinear and the experts can specialise on top of a common, informative feature space rather than each having to relearn low level feature interactions. Because the embedding is shared, every expert consumes an identically shaped representation, which is what allows experts with radically different internal structure to be mixed by a common router. A router specific summary $\bar{\mathbf{z}} \in \mathbb{R}^d$ is obtained by global average pooling over the feature axis, $\bar{\mathbf{z}} = \frac{1}{F}\sum_{i=1}^{F} \mathbf{Z}_{i,:}$, which gives the router a compact, permutation stable view of the flow on which to base its routing decision.

### 3.8.3. Heterogeneous Expert Pool

The expert pool comprises eight architecturally heterogeneous sub networks, each genuinely deep but individually lightweight, so that different experts can specialise on different structural regularities in the traffic. Heterogeneity is a deliberate departure from the common practice of instantiating many copies of a single expert architecture; the intuition, validated by the architectural ablation of Section 3.12.1, is that a diverse pool offers the router a richer set of computational primitives to compose, so that volumetric floods, low rate stealth attacks and scanning behaviours can each be served by the expert whose inductive bias best matches them. Table 3.4 enumerates the eight experts together with their parameter counts and standalone floating point operation counts, drawn from the per expert cost profile of a representative binary configuration of the proposed model, and the inductive bias each contributes.

Table 3.4: Heterogeneous expert pool and resource profile

| Idx | Expert | Params | FLOPs | Inductive bias |
|---|---|---|---|---|
| 0 | MLP | 8 992 | 17 952 | General purpose nonlinear classifier |
| 1 | Wide MLP | 26 944 | 53 824 | High capacity for complex nonlinear feature interactions |
| 2 | CNN | 1 600 | 107 264 | Local feature patterns via 1D convolution with batch normalisation |
| 3 | Multiscale CNN | 4 672 | 265 936 | Parallel 3/5/7 kernels for multi receptive field patterns |
| 4 | Efficient attention | 1 360 | 79 816 | Cheap single head attention for long range dependencies |
| 5 | Autoencoder | 18 248 | 37 545 | Bottleneck reconstruction for anomaly oriented representation |
| 6 | Attention MLP | 46 290 | 92 530 | Squeeze and excite style per feature gating then MLP |
| 7 | ResMLP | 9 296 | 18 644 | Residual MLP with layer normalisation for deeper tabular learning |

Each expert $e_j$ maps the shared embedding to a fixed size representation $\mathbf{u}_j = e_j(\mathbf{Z}) \in \mathbb{R}^m$, so that regardless of internal architecture all experts emit comparably shaped outputs that the router can mix. The internal design of each expert is chosen to endow it with a distinct inductive bias. The multilayer perceptron expert is a two layer dense stack with dropout that serves as a general purpose nonlinear classifier and a cheap default. The wide multilayer perceptron expert widens the hidden layer by a factor of three before projecting back, trading parameters for the

capacity to represent complex nonlinear feature interactions that the narrow expert cannot. The convolutional expert applies two one dimensional convolutions with batch normalisation and global average pooling, which biases it toward local patterns among adjacent feature positions. The multiscale convolutional expert runs parallel convolutions with kernels of width three, five and seven and concatenates their responses, so that it captures patterns at several receptive field scales simultaneously, a bias well suited to attacks whose signatures span feature groups of different sizes. The efficient attention expert implements a single head attention mechanism that models long range dependencies among feature positions at a cost far below full multi head attention. The autoencoder expert compresses the embedding through a bottleneck and reconstructs it under an auxiliary reconstruction objective, which biases it toward an anomaly oriented representation that is sensitive to deviations from the reconstructed manifold. The attention multilayer perceptron expert applies a squeeze and excite style per feature gating, in which a small network learns to reweight each feature channel before a dense stack, giving it a bias toward selectively amplifying the most discriminative features. The residual multilayer perceptron expert stacks dense blocks with residual connections and layer normalisation, which stabilises optimisation and permits greater effective depth for tabular feature learning. This deliberate spread of biases, from local convolution to global attention to reconstruction to feature gating, is what allows the router to compose a response tailored to each flow. The pool originally included two full transformer experts, a compact transformer encoder block and a per feature token attention expert, but the routing analysis of Section 3.11 found that the router assigned both essentially zero weight for every attack class while the floating point profiler flagged them as consistently among the most expensive experts to evaluate, owing to their large multi head attention matrix multiplications. They were therefore removed from the default pool, a decision that is entirely consistent with the lightweight objective and that was made on the basis of measured routing behaviour rather than assumption.

### 3.8.4. Noisy Top-$k$ Router and Sparse Gating

The router is a lightweight two layer multilayer perceptron that produces noisy top-$k$ gate logits, following the noisy top-$k$ gating principle. A single dense gate is under powered once the expert

pool grows beyond a handful of experts, so a small two layer router is used as the best balance of simplicity and routing quality at this scale. Given the pooled embedding $\bar{\mathbf{z}}$, the router computes a clean logit vector and a data dependent noise scale,

$$\mathbf{h} = \sigma(\mathbf{W}_r \bar{\mathbf{z}}), \tag{3.3}$$

$$\boldsymbol{\ell} = \mathbf{W}_g \mathbf{h}, \qquad \mathbf{s} = \mathrm{softplus}(\mathbf{W}_n \mathbf{h}), \tag{3.4}$$

and, during training only, perturbs the logits with heteroscedastic Gaussian noise to encourage exploration and to prevent the router from collapsing prematurely onto a fixed subset of experts,

$$\tilde{\boldsymbol{\ell}} = \boldsymbol{\ell} + \boldsymbol{\varepsilon} \odot \mathbf{s}\,\eta, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{3.5}$$

where $\eta$ is the noise standard deviation and $\odot$ is the elementwise product; at inference the clean logits are used. The $k$ largest noisy logits are then selected, a softmax is applied restricted to the selected experts, and the resulting weights are scattered back into a dense but sparse gate vector $\mathbf{a} \in \mathbb{R}^N$ whose support is exactly the $k$ selected experts,

$$(\mathbf{v}, \mathcal{S}) = \text{top-}k(\tilde{\boldsymbol{\ell}}), \qquad a_j = \begin{cases} \dfrac{\exp(v_j)}{\sum_{j' \in \mathcal{S}} \exp(v_{j'})}, & j \in \mathcal{S}, \\ 0, & j \notin \mathcal{S}. \end{cases} \tag{3.6}$$

The default configuration uses $k = 3$, consistent with the guidance of top-2 to top-3 routing for an eight to ten expert pool, so that only three of the eight experts are evaluated for any given flow. Restricting the softmax to the selected experts, rather than softmaxing over all experts and then masking, ensures that the active gate weights sum to one and that the deselected experts contribute exactly nothing to the forward pass and, crucially, nothing to the per sample compute.

### 3.8.5. Load Balancing Auxiliary Objective

Sparse routers are prone to a degenerate equilibrium in which the router learns to send almost every flow to a small clique of experts, leaving the remaining experts undertrained and wasting the capacity that motivated the diverse pool in the first place. To counteract this, a load balancing

auxiliary loss penalises uneven expert utilisation across a batch. For a batch of size $B$ with sparse gate vectors $\mathbf{a}^{(1)}, \ldots, \mathbf{a}^{(B)}$, the per expert importance is the total gate mass each expert receives, $I_j = \sum_{b=1}^{B} a_j^{(b)}$, and the auxiliary loss is the squared coefficient of variation of the importance vector,

$$\mathcal{L}_{\mathrm{lb}} = \lambda \, \frac{\mathrm{Var}(\mathbf{I})}{\left(\bar{\mathbf{I}}\right)^2 + \epsilon}, \tag{3.7}$$

with balancing weight $\lambda$ and a small constant $\epsilon$ for numerical stability. Minimising the squared coefficient of variation drives the importance toward uniformity across experts without dictating which expert should serve which flow, so the router remains free to specialise while every expert receives enough gradient signal to remain useful. This auxiliary term is added to the primary classification loss, and its modest weight ensures that it regularises routing without overwhelming the detection objective. The effect of this objective is directly observable in the expert utilisation heatmaps of Chapter Four, which show that all eight experts receive non trivial gate mass rather than a few experts monopolising the routing.

### 3.8.6. Fusion Head and Classifier

The sparsely weighted expert outputs are combined by the gate weighted sum

$$\mathbf{u} = \sum_{j=1}^{N} a_j \, \mathbf{u}_j = \sum_{j \in \mathcal{S}} a_j \, e_j(\mathbf{Z}), \tag{3.8}$$

in which the second equality makes explicit that only the $k$ selected experts contribute. The mixed representation is passed through a two layer fusion head with dropout, which combines the possibly complementary views of the selected experts into a single discriminative representation before the final softmax classifier,

$$f_\theta(\mathbf{X}) = \mathrm{softmax}\big(\mathbf{W}_c \, h_\omega(\mathbf{u})\big). \tag{3.9}$$

A two layer fusion head, rather than a single linear layer, is used so that the model can learn nonlinear interactions between the outputs of the co selected experts rather than merely averaging them, which matters when the router selects experts with complementary rather than redundant

biases. Algorithm 1 presents the complete forward pass.

**Algorithm 1** Forward pass of the proposed sparse MoE detector

**Require:** flow tensor $\mathbf{X} \in \mathbb{R}^{F\times 1}$, experts $\{e_j\}_{j=1}^{N}$, sparsity $k$, training flag $\tau$
**Ensure:** class distribution $\hat{\mathbf{p}} \in \Delta^{C-1}$

$\mathbf{Z} \leftarrow g_\phi(\mathbf{X})$ ▷ shared two layer embedding, Eq. (3.2)
$\bar{\mathbf{z}} \leftarrow \mathrm{GlobalAvgPool}(\mathbf{Z})$ ▷ compact router input
$\mathbf{h} \leftarrow \sigma(\mathbf{W}_r \bar{\mathbf{z}})$; $\quad \boldsymbol{\ell} \leftarrow \mathbf{W}_g \mathbf{h}$; $\quad \mathbf{s} \leftarrow \mathrm{softplus}(\mathbf{W}_n \mathbf{h})$
**if** $\tau$ is true **then** ▷ inject exploration noise during training only
    $\tilde{\boldsymbol{\ell}} \leftarrow \boldsymbol{\ell} + \boldsymbol{\varepsilon} \odot \mathbf{s}\,\eta,\ \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
**else**
    $\tilde{\boldsymbol{\ell}} \leftarrow \boldsymbol{\ell}$
**end if**
$(\mathbf{v}, \mathcal{S}) \leftarrow \text{top-}k(\tilde{\boldsymbol{\ell}})$; $\quad$ compute sparse gates $\mathbf{a}$ by Eq. (3.6)
**if** $\tau$ is true **then**
    add $\mathcal{L}_{\mathrm{lb}} = \lambda\, \mathrm{Var}(\mathbf{I}) / (\bar{\mathbf{I}}^2 + \epsilon)$ to the loss ▷ Eq. (3.7)
**end if**
$\mathbf{u} \leftarrow \sum_{j\in\mathcal{S}} a_j\, e_j(\mathbf{Z})$ ▷ only $k$ experts evaluated, Eq. (3.8)
$\hat{\mathbf{p}} \leftarrow \mathrm{softmax}(\mathbf{W}_c\, h_\omega(\mathbf{u}))$
**return** $\hat{\mathbf{p}}$

### 3.8.7. Lightweight Configuration for the Edge

The architecture is rendered lightweight through a set of deliberate width reductions targeted at the components that the floating point profiler identified as most expensive. The internal width of each expert and the shared embedding dimension were halved from thirty two to sixteen, the router hidden width and the fusion head were shrunk to match, and the bottleneck ratio of the attention multilayer perceptron expert was tightened. Because the dominant cost of the attention, convolutional and attention multilayer perceptron experts scales with the embedding dimension, and for the per feature gating of the attention multilayer perceptron expert with its square, this halving compounds into a reduction in per expert floating point operations substantially larger than the reduction in parameter count alone would suggest, and it targets precisely the experts that the profiler had flagged as costly. The combination of width reduction with top-$k$ sparsity is what keeps the model lightweight in the sense that matters for the edge: the number of parameters that must actually be evaluated for any single prediction, and the associated floating point operations, are only those of the $k$ selected experts plus the shared embedding, router and fusion head, rather than those of the whole pool. This per prediction cost is measured

directly in Chapter Four through a per prediction routing inspector. The next section specifies the comparator models against which this framework is evaluated.

### 3.9. Baseline Models

To situate the proposed framework, seven baseline detectors spanning four modelling families are trained and evaluated under the identical preprocessing, partitioning and metric protocol. The classical machine learning family comprises a Random Forest with one hundred and fifty trees of maximum depth fourteen, an XGBoost gradient boosted ensemble with one hundred and fifty trees of maximum depth six and a learning rate of fifteen hundredths, and a radial basis Support Vector Machine trained on a seeded subsample for tractable runtime. The deep learning family comprises a one dimensional convolutional network and a long short term memory recurrent network. The hybrid family comprises a convolutional long short term memory network that stacks convolutional feature extraction beneath recurrent temporal modelling. The transformer family comprises a lightweight transformer encoder. All seven baselines consume the same scaled and oversampled training tensors as the proposed model, and the four sequence consuming baselines consume the same pseudo sequential representation defined in Section 3.7.6. Table 3.5 summarises the baselines and their principal configuration.

Table 3.5: Baseline detectors and configurations

| | | |
|---|---|---|
| Deep learning | CNN | 1D convolutional network on the pseudo sequence |
| Deep learning | LSTM | Recurrent network on the pseudo sequence |
| Hybrid | CNN-LSTM | Convolutional feature extraction beneath recurrent modelling |
| Transformer | Lightweight Transformer | Compact transformer encoder on the pseudo sequence |

The rationale for this breadth is that each family carries a different inductive bias and a different computational profile, so that the comparison probes not only whether the proposed model is accurate but whether its particular combination of accuracy and efficiency is genuinely distinctive; the deep and hybrid sequence models test whether sequence oriented inductive bias helps or hinders on shuffled tabular flow features; and the transformer provides a modern attention based reference whose cost the lightweight design seeks to undercut. The training

protocol shared by all neural models is specified next.

### 3.10. Optimisation, Training and Validation Protocol

All neural models, including the proposed sparse MoE, are trained under an identical protocol so that architectural differences rather than training differences drive the comparison. Optimisation uses the Adam optimiser with an initial learning rate of five thousandths and the sparse categorical cross entropy loss, to which, for the proposed model, the load balancing auxiliary loss of Equation (3.7) is added. Training runs for up to one thousand epochs with a batch size of two thousand and forty eight, but is governed by two callbacks that make the effective training length adaptive. Early stopping monitors the validation loss with a patience of fifteen epochs and restores the best weights, so that training halts once the validation loss ceases to improve and the model is never evaluated in an overfitted state. Learning rate reduction on plateau halves the learning rate after four epochs without validation improvement, down to a floor of one millionth, so that the optimiser can settle into sharp minima once coarse progress stalls. Table 3.6 consolidates the optimisation hyperparameters.

Table 3.6: Training and optimisation hyperparameters

| **Hyperparameter** | **Value** |
|---|---|
| Optimiser | Adam |
| Initial learning rate | $5 \times 10^{-3}$ |
| Loss | Sparse categorical cross entropy (plus load balancing loss for the MoE) |
| Maximum epochs | 1000 |
| Batch size | 2048 |
| Early stopping | Monitor validation loss, patience 15, restore best weights |
| Learning rate schedule | Halve on plateau, patience 4, floor $1 \times 10^{-6}$ |
| Router sparsity $k$ | 3 (framework default) |
| Expert width / embedding $d$ | 16 (lightweight configuration) |
| Load balancing weight $\lambda$ | $5 \times 10^{-3}$ |

The validation subset serves a dual role. It drives the early stopping and learning rate schedule, and it provides an unbiased checkpoint selection criterion, since the restored weights are those that minimised the validation loss rather than the training loss. The test subset is touched only once, at final evaluation, and never influences training, checkpointing or hyperparameter choices, which preserves it as a faithful estimate of generalisation. The training and validation

loss and accuracy trajectories of every neural model are recorded and exported as training curves, illustrated for the proposed model on the IoT-23 binary task in Figure 3.3, and these curves are used in Chapter Four to diagnose convergence behaviour and to confirm the absence of overfitting.

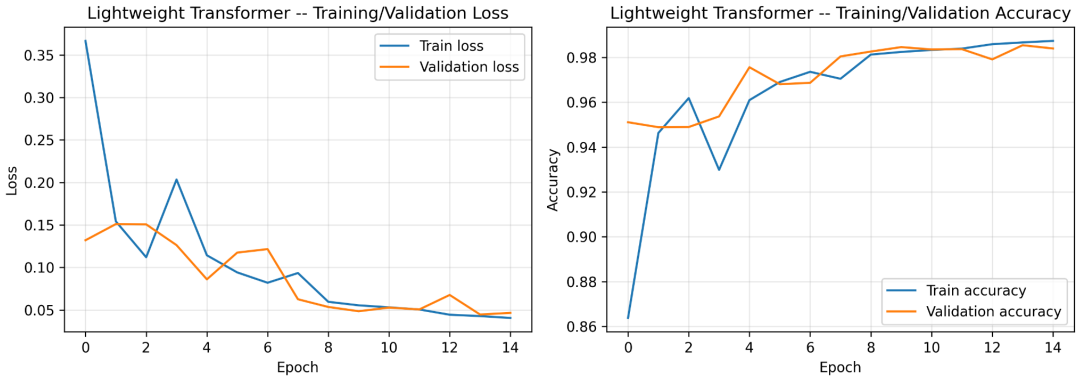



Figure 3.3: Training and validation curves for IoT-23 binary detection

### 3.11. Explainability Framework

Trustworthy intrusion detection requires that a model's decisions be inspectable, and the framework is therefore instrumented for explainability at four complementary levels, all applicable without modification to both the binary and the multiclass tasks. The first level is routing level explainability, which is native to the Mixture of Experts structure. An expert utilisation analysis records, for each class, the average gate mass assigned to each expert, producing a utilisation heatmap that reveals whether routing decisions are semantically meaningful, that is, whether different attack families are served by different experts. This routing level view is complemented by a per prediction inspector that, for any individual flow, reports exactly which experts the router activated, their gate weights and each active expert's standalone parameter count and floating point operations, so that the actual compute cost of a single prediction can be checked against a target device's budget rather than only the whole model's cost.

The second level is global, model agnostic explainability by permutation feature importance computed on the trained model, which quantifies how much test performance degrades when each feature is independently shuffled, thereby ranking features by their contribution to the model's decisions without reference to the model's internals. The third level is prediction level

attribution by KernelSHAP, a model agnostic Shapley value estimator that operates directly on the model's prediction function against a small representative background sample and a subsampled explanation set to keep runtime tractable. KernelSHAP yields both global summary plots, per class for the multiclass task and for the attack class in the binary task, and a local waterfall plot decomposing an individual prediction into its per feature contributions. A fast exact TreeSHAP analysis of the Random Forest baseline provides a reference point against which the model agnostic attributions can be sanity checked. The fourth level is local surrogate explainability by LIME, which fits an interpretable linear surrogate in the neighbourhood of an individual flow and is used to contrast the feature contributions behind a correctly classified flow against those behind a misclassified one. Together these four levels span routing, global feature, and individual prediction explanations, and their outputs are interpreted in Chapter Four. The framework's robustness protocol is described next.

### 3.12. Robustness Evaluation by Withheld Attack Families

Beyond in distribution accuracy, an intrusion detector must retain some ability to flag attacks it has never seen, since novel campaigns appear continually. Robustness to novelty is therefore assessed by a simulated zero day protocol that withholds one entire attack family from the training data, retrains the model on the remaining families and normal traffic, and then measures how well the trained model separates the withheld family from normal traffic at inference using its predictive confidence as an anomaly signal. Because the withheld family is absent from training, its flows are presented to the model as genuinely novel. At inference the maximum softmax confidence is used as a novelty score, on the premise that flows the model cannot confidently assign to any known class are candidate novelties, and the lowest confidence quartile is flagged as novel. The detection recall and flagging precision on the withheld family are then reported. This protocol is meaningful only when multiple attack families exist, so it is exercised on the multiclass configurations. It provides a conservative, threshold based lower bound on the framework's novelty sensitivity rather than a bespoke novelty detector, and it is interpreted in that light in Chapter Four. The ablation designs that isolate the framework's individual design choices are specified next.

### 3.12.1. Architectural Ablation Experiment

The architectural ablation retrains the framework under a family of controlled variants and evaluates each on the same held out test set, so that the effect of each design axis can be read off directly. Four axes are varied. The sparsity axis contrasts top-1, top-2, top-3 and top-4 routing against dense routing in which all experts are active. The expert count axis contrasts pools of four, eight and ten experts. The heterogeneity axis contrasts the diverse pool against a homogeneous pool of identical multilayer perceptron experts. For each variant the macro F1 score, accuracy, parameter count and floating point operations are recorded, so that accuracy can be traded off against cost. This design allows the study to answer whether sparsity, diversity and pool size each contribute, and at what computational price, and its results are reported in Chapter Four.

## 3.13. Evaluation Metrics

Detection quality and computational efficiency are quantified by two complementary batteries of metrics, computed identically for the binary and multiclass tasks. The detection battery reports accuracy, macro averaged and weighted precision, recall and F1, and balanced accuracy, which averages per class recall and is robust to the class imbalance present in the benchmark corpora. It reports the Matthews correlation coefficient and Cohen's kappa as chance corrected agreement measures that penalise trivial majority prediction. From the confusion matrix it derives class averaged false positive rate, false negative rate and specificity, and a detection rate defined as recall on attack traffic treating the normal class as negative and all attack classes as positive. From the predicted probabilities it reports the area under the receiver operating characteristic curve, the area under the precision recall curve, the log loss and the Brier score, the latter two quantifying probability calibration. The efficiency battery reports the parameter count, the model size in megabytes at single precision, the per sample inference latency in milliseconds and the floating point operations per inference. Formally, with true positives, false positives, true negatives and false negatives denoted $TP$, $FP$, $TN$ and $FN$, the core rates are

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1} = \frac{2\,\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{3.10}$$

and the Matthews correlation coefficient is

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \tag{3.11}$$

Macro averaging weights every class equally regardless of frequency and is therefore the primary lens for the imbalanced multiclass tasks, whereas weighted averaging and raw accuracy are dominated by the frequent classes and are reported for completeness and comparability. Reporting calibration alongside discrimination is deliberate, because a deployed detector that is confidently wrong is more dangerous than one whose uncertainty is well calibrated. The complexity accounting that grounds the efficiency claims is described next.

### 3.14. Computational Complexity Accounting

The efficiency claims of the framework rest on an explicit accounting of both the static and the dynamic computational cost of the Mixture of Experts. The static cost is the total parameter count and the standalone floating point operations of every component, including each of the eight experts individually, which yields the per expert price list of Table 3.4. The dynamic cost is the cost actually incurred per prediction, which under top-$k$ routing is only the cost of the shared embedding, the router, the fusion head and the $k$ selected experts, rather than the full pool. For a pool of $N$ experts with per expert cost $c_j$, the expected per sample expert cost under a routing distribution that selects set $\mathcal{S}$ is $\sum_{j \in \mathcal{S}} c_j$ with $|\mathcal{S}| = k$, in contrast to $\sum_{j=1}^{N} c_j$ for a dense model, so that sparsity converts total capacity into a fraction of the compute a dense model of equal capacity would demand. This distinction between total and active cost is the quantitative heart of the lightweight claim, and it is measured directly in Chapter Four both in aggregate, through the expert cost breakdown, and per prediction, through the routing inspector that reports the active parameter fraction of individual flows. The chapter concludes by consolidating these methodological elements.

## 3.15. Chapter Summary

This chapter has specified, in reproducible detail, the materials and methods of the study. It established an experimental, design science research design with a factorial structure of two datasets crossed with binary and multiclass tasks, and it fixed a deterministic computational environment governed by a single global seed and operation level determinism. It documented the two benchmark corpora, IoT-23 and Edge-IIoTset, and developed a leakage aware preprocessing pipeline comprising the removal of uninformative, identifying and target leaking attributes, per class Tukey winsorisation, stratified partitioning, training split only scaling and minority only synthetic oversampling. It formalised the proposed lightweight sparse Mixture of Experts framework, comprising a shared two layer embedding, a pool of eight heterogeneous experts, a noisy top-$k$ router regularised by a load balancing auxiliary objective and a two layer fusion head, and it explained the lightweight configuration that renders the model suitable for the edge. It specified the seven baselines, the shared optimisation and validation protocol, the four level explainability framework, the withheld family robustness protocol, the architectural ablation design, the detection and efficiency metrics, and the static and dynamic complexity accounting. The empirical results produced by instantiating this methodology across all four primary configurations, together with the ablations, the routing analysis, the explainability outputs and the robustness evaluation, are presented and critically discussed in the next chapter.

# CHAPTER FOUR

# RESULTS AND DISCUSSION

## 4.1. Introduction to the Chapter

This chapter reports and critically interprets the empirical results produced by instantiating the methods of Chapter Three across all four primary experimental configurations, together with the architectural ablation, the routing analysis, the four level explainability outputs and the robustness evaluation. Every numerical value reported here is drawn from the experimental artefacts generated by the pipeline, and every figure referenced corresponds to a plot exported by the notebooks. The chapter opens by cataloguing the experimental outputs into logical categories. It then examines the preprocessing diagnostics that establish the integrity of the data on which the models were trained. It proceeds dataset by dataset through the comparative detection results, contrasting the proposed sparse Mixture of Experts framework against the seven baselines on both detection quality and computational efficiency, and drawing on the per class classification reports, confusion matrices and training curves to explain the observed behaviour rather than merely to restate it. A cross dataset synthesis then distils the findings that generalise across both corpora from those that are dataset specific. The chapter next analyses computational efficiency in depth, using the per expert cost breakdown and the per prediction routing inspector to substantiate the lightweight claim, before turning to the expert utilisation analysis that reveals the semantics of the router's decisions. The explainability results are then interpreted at the global feature and individual prediction levels, followed by the robustness evaluation on withheld attack families. The architectural ablation quantifies the contribution of sparsity, diversity and pool size. Throughout, the findings are related to the prior literature so that the study's contributions and points of departure are made explicit. The chapter closes with a consolidating summary.

## 4.2. Catalogue of Experimental Outputs

The experiments generated a large and structured body of artefacts, which are organised here into the categories that frame the remainder of the chapter. The detection performance category comprises, for each of the four primary configurations, a model comparison table reporting the full detection and efficiency battery for all eight models, a per class classification report for the proposed model, and a confusion matrix for every model. The convergence category comprises training and validation loss and accuracy curves for every neural model. The computational performance category comprises a per expert static cost breakdown and a per prediction routing inspection for the proposed model, together with the parameter, latency and floating point counts embedded in the comparison tables. The routing category comprises an expert utilisation heatmap and utilisation matrix for every configuration. The explainability category comprises permutation importance rankings, global and local KernelSHAP plots, per class SHAP summaries for the multiclass tasks, and LIME explanations contrasting correctly classified and misclassified flows. The ablation experiment comprises the architectural ablation table. The preprocessing category comprises class distribution plots, per feature outlier summaries and before and after capping distributions. This catalogue confirms that the interpretation that follows is grounded in a complete and traceable set of experimental outputs.

## 4.3. Preprocessing Diagnostics

Before the detection results are examined, the integrity of the preprocessing is verified, because the validity of every downstream metric depends on the data having been cleaned, tempered and balanced as intended. The outlier analysis confirms the heavy-tailed nature of the raw flow statistics: on IoT-23, for example, the originator IP byte and originator packet attributes each flag roughly one flow in six as lying beyond the Tukey fence, and the flow duration flags roughly one flow in eight, whereas the responder-side counters flag well under one percent, which shows that the extreme values are concentrated in a small number of high-volume attributes rather than being diffuse. Figure 4.1 shows the per attribute boxplots that motivate the winsorisation.

The class distribution diagnostics confirm the pronounced imbalance across both corpora

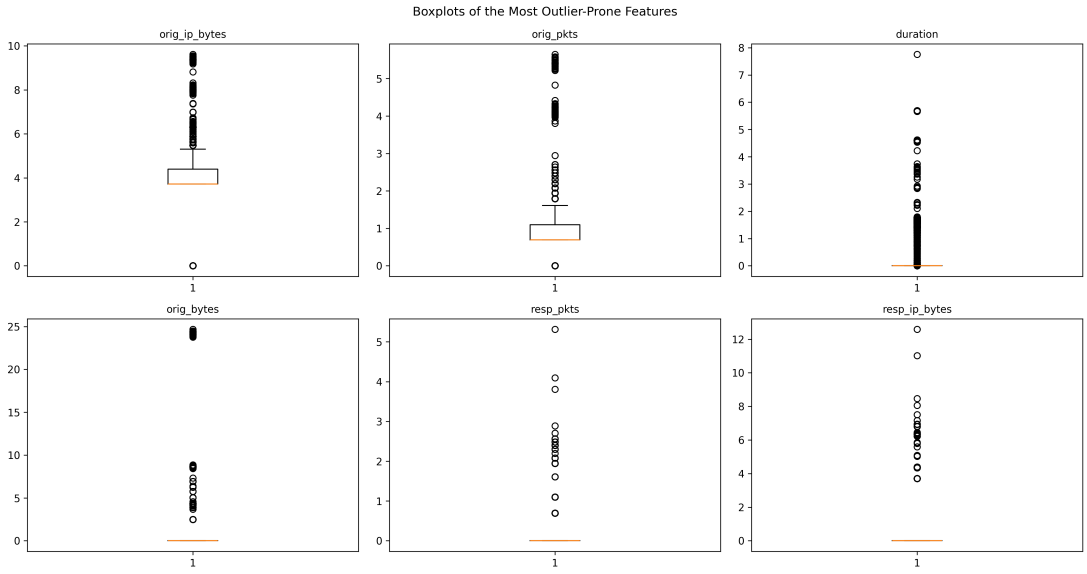



Figure 4.1: IoT-23 binary outlier boxplots

and thereby justify the reliance on macro averaged and balanced metrics and the minority only oversampling. In the Edge-IIoTset multiclass configuration, illustrated in Figure 4.3, the man in the middle and fingerprinting families comprise only a few tens to a few hundred flows against hundreds of thousands of normal and distributed denial of service flows, an imbalance of several orders of magnitude, which foreshadows the finding that macro F1 on Edge-IIoTset multiclass is governed almost entirely by the model's handling of these rarest families. The IoT-23 multiclass distribution in Figure 4.2 is less extreme but still markedly skewed toward benign and scanning traffic, which motivates the minority only oversampling applied to its command and control and botnet families.

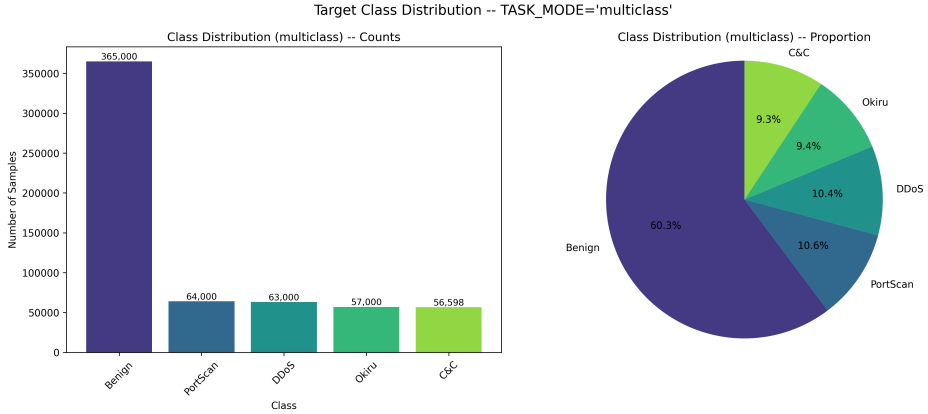



Figure 4.2: IoT-23 multiclass class distribution

With the preprocessing integrity established, the chapter turns to the comparative detection

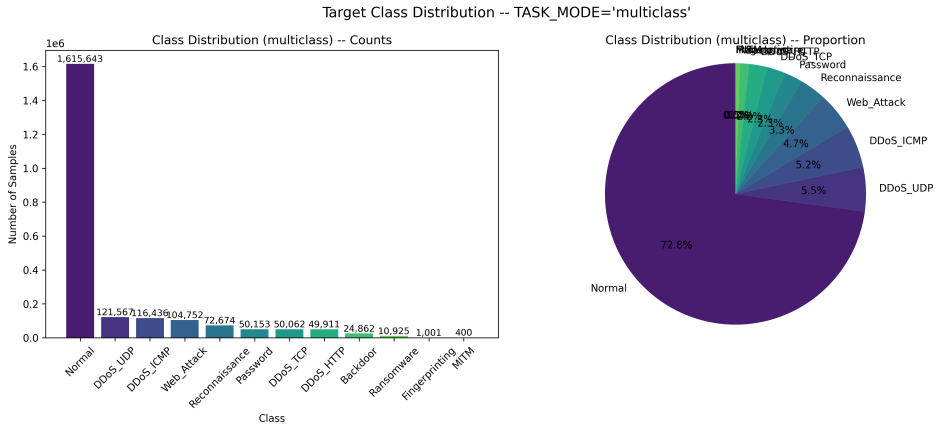



Figure 4.3: Edge-IIoTset multiclass class distribution

results.

## 4.4. Comparative Detection Performance

This section reports the comparative detection results dataset by dataset. For each configuration a comparison table reports accuracy, macro F1, weighted F1 and the Matthews correlation coefficient as percentages, alongside the parameter count, per sample latency and floating point operations, for the proposed model and all seven baselines. The proposed model's per class behaviour is then examined through its classification report and confusion matrix, and its convergence through its training curves, so that the aggregate metrics are explained rather than merely tabulated.

### 4.4.1. IoT-23 Binary Detection

On the IoT-23 binary task the proposed framework attains 99.88 percent accuracy, a macro F1 of 99.88 percent and a Matthews coefficient of 99.75 percent, as reported in Table 4.1. It is marginally behind the Random Forest and XGBoost, which reach approximately 99.93 percent, and marginally ahead of the convolutional network, while the recurrent, hybrid and transformer baselines fall away sharply. The long short term memory network is the striking failure here, at only 60.86 percent accuracy and a near zero Matthews coefficient of 9.39 percent, which indicates that it degenerated to near majority prediction on this corpus.

The per class report of the proposed model shows balanced precision and recall on both the

Table 4.1: IoT-23 binary detection results

| Model | Acc | F1 | wF1 | MCC | Params | Lat. ms | FLOPs |
|---|---|---|---|---|---|---|---|
| Random Forest | 99.93 | 99.93 | 99.93 | 99.85 | 34 372 | – | – |
| XGBoost | 99.92 | 99.92 | 99.92 | 99.83 | 5 236 | – | – |
| **Proposed Sparse MoE** | **99.88** | **99.88** | **99.88** | **99.75** | 170 055 | 93.21 | 953 989 |
| CNN | 99.87 | 99.87 | 99.87 | 99.73 | 25 186 | 87.36 | 575 180 |
| SVM | 98.03 | 97.93 | 98.02 | 95.89 | 630 | – | – |
| CNN-LSTM Hybrid | 88.26 | 87.11 | 87.90 | 76.02 | 39 586 | 91.29 | 289 932 |
| Lightweight Transformer | 87.23 | 85.73 | 86.68 | 74.51 | 29 794 | 87.34 | 3 396 208 |
| LSTM | 60.86 | 39.22 | 46.67 | 9.39 | 21 090 | 91.09 | 2 220 |

normal and attack classes at or above 99.8 percent, and the confusion matrix confirms that both false alarms and missed attacks are rare. The contrast between the proposed model's robustness and the collapse of the recurrent and hybrid baselines is instructive: because the pseudo sequence axis is a feature axis rather than a temporal axis, purely recurrent processing has no genuine temporal structure to exploit and can be actively harmful, whereas the proposed model's router can simply route these flows to its multilayer perceptron and convolutional experts, whose inductive biases suit tabular flow features, and thereby avoid the recurrent failure mode entirely. This is a concrete illustration of the advantage of conditional computation over a heterogeneous pool: the model is not committed to a single inductive bias that may be mismatched to the data.

### 4.4.2. IoT-23 Multiclass Detection

On the five class IoT-23 multiclass task the proposed framework attains 99.89 percent accuracy, a macro F1 of 99.87 percent and a Matthews coefficient of 99.82 percent, as shown in Table 4.2, trailing only the Random Forest and XGBoost by a few hundredths of a percentage point. The deep and hybrid baselines fail dramatically on this task: the convolutional network, the recurrent network and the hybrid all fall to between 59 and 63 percent accuracy with macro F1 scores between 17 and 32 percent, indicating near total collapse on the minority attack families. The proposed model's per class report, by contrast, shows F1 scores at or above 99.4 percent for every one of the five classes, including the command and control family that the weaker models confuse with benign traffic.

The dramatic separation between the proposed model and the deep baselines on this task is the single most compelling demonstration of the value of the heterogeneous expert design. The

Table 4.2: IoT-23 multiclass detection results

| Model | Acc | F1 | wF1 | MCC | Params | Lat. ms | FLOPs |
|---|---|---|---|---|---|---|---|
| Random Forest | 99.93 | 99.91 | 99.93 | 99.88 | 37 680 | – | – |
| XGBoost | 99.92 | 99.90 | 99.92 | 99.87 | 17 634 | – | – |
| **Proposed Sparse MoE** | **99.89** | **99.87** | **99.89** | **99.82** | 170 106 | 100.58 | 954 103 |
| SVM | 99.69 | 99.61 | 99.69 | 99.48 | 430 | – | – |
| Lightweight Transformer | 98.05 | 96.55 | 98.04 | 96.74 | 29 893 | 93.36 | 3 396 418 |
| LSTM | 63.04 | 26.71 | 51.75 | 22.04 | 21 189 | 94.48 | 2 430 |
| CNN | 60.85 | 17.65 | 47.43 | 12.29 | 25 285 | 96.11 | 575 390 |
| CNN-LSTM Hybrid | 59.48 | 31.75 | 56.36 | 35.11 | 39 685 | 96.24 | 290 142 |

confusion matrix of the convolutional baseline, shown for contrast in Figure 4.5, reveals that it collapses the Okiru and port scan families entirely into other classes, whereas the proposed model, by routing each family to the experts best suited to it, preserves per class competence across the taxonomy. This is direct empirical support for the thesis that architectural diversity, orchestrated by a learned router, generalises across attack families in a way that a single monolithic deep network does not.

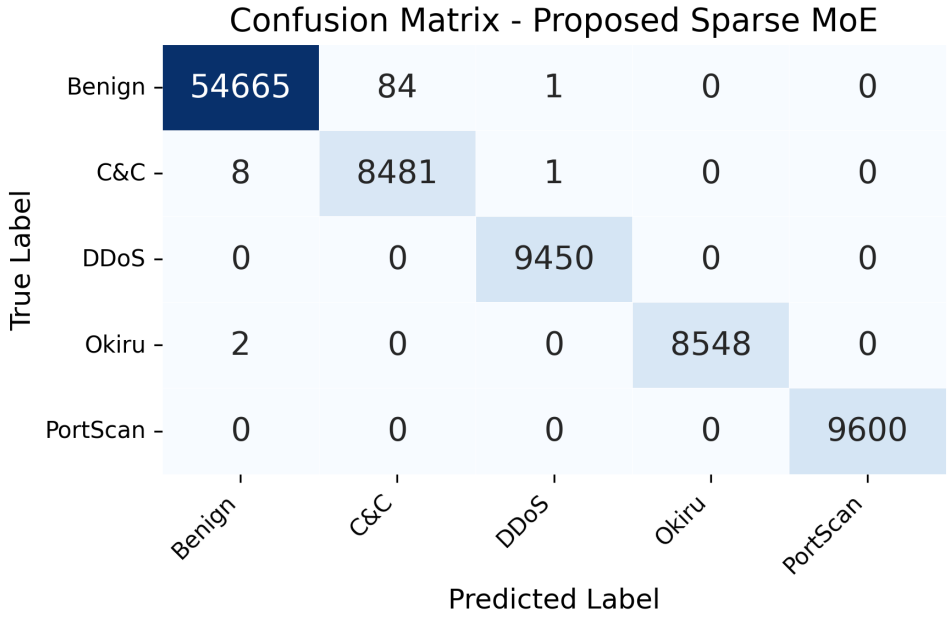



Figure 4.4: Proposed sparse MoE confusion matrix on IoT-23 multiclass task

Table 4.3 reports the per class performance of the proposed model, and the numbers make the contrast with the deep baselines stark. The proposed model classifies the distributed denial of service, Okiru and port scan families at essentially perfect F1 and recognises the command and control family, which the convolutional and recurrent baselines confuse with benign traffic,

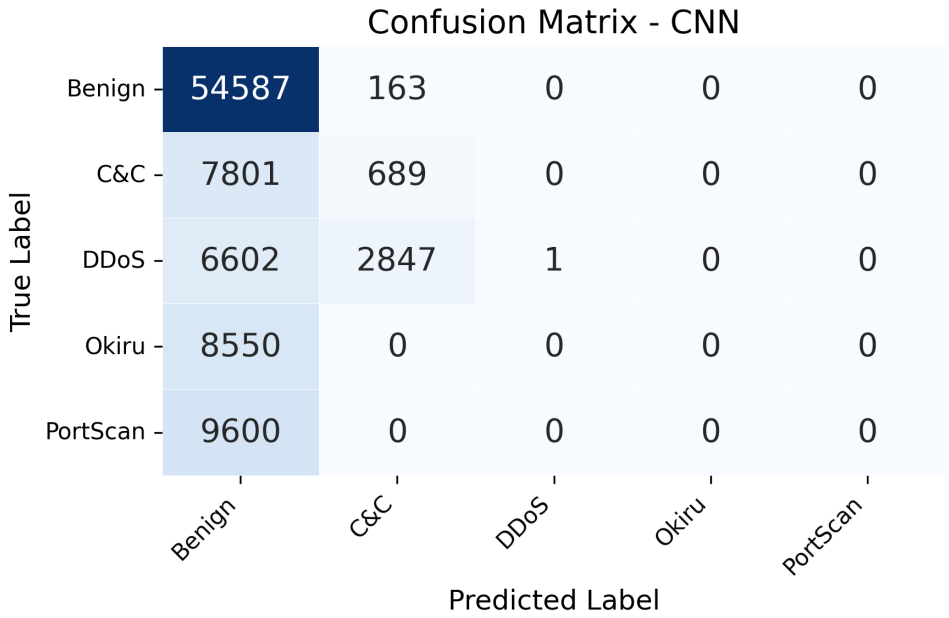



Figure 4.5: CNN confusion matrix on IoT-23 multiclass task

at 99.45 percent F1. The command and control family is the discriminating case: it is the family on which the weak baselines collapse, because its flow statistics overlap with benign traffic and only a model that can dedicate specialised competence to it, rather than averaging it into a single decision boundary, can separate it reliably. That the proposed model achieves near perfect separation here, while the convolutional network's macro F1 on the whole task falls to 17.65 percent, is the sharpest single illustration in the study of the generalisation benefit of heterogeneous conditional computation on imbalanced attack taxonomies.

Table 4.3: Proposed sparse MoE per-class performance on IoT-23

| **Class** | **Precision** | **Recall** | **F1** | **Support** |
|---|---|---|---|---|
| Benign | 99.98 | 99.84 | 99.91 | 54 750 |
| Command and control | 99.02 | 99.89 | 99.45 | 8 490 |
| DDoS | 99.98 | 100.00 | 99.99 | 9 450 |
| Okiru | 100.00 | 99.98 | 99.99 | 8 550 |
| Port scan | 100.00 | 100.00 | 100.00 | 9 600 |

### 4.4.3. Edge-IIoTset Binary Detection

The Edge-IIoTset binary task is the easiest of the four configurations, and the proposed framework attains a perfect 100.00 percent on every detection metric, as reported in Table 4.4. It shares this ceiling with the Random Forest, XGBoost, the convolutional network, the recurrent network and

the hybrid, while the support vector and transformer baselines fall only fractions of a percentage point short. The fact that so many models saturate this task indicates that the normal and attack populations in Edge-IIoTset are almost linearly separable after preprocessing, so the task discriminates between models principally on efficiency rather than accuracy, a point developed in Section 4.6.

Table 4.4: Edge-IIoTset binary detection results

| Model | Acc | F1 | wF1 | MCC | Params | Lat. ms | FLOPs |
|---|---|---|---|---|---|---|---|
| **Proposed Sparse MoE** | **100.00** | **100.00** | **100.00** | **100.00** | 251 670 | 83.70 | 1 331 854 |
| Random Forest | 100.00 | 100.00 | 100.00 | 100.00 | 1 264 | – | – |
| XGBoost | 100.00 | 100.00 | 100.00 | 100.00 | 332 | – | – |
| CNN | 100.00 | 100.00 | 100.00 | 100.00 | 25 186 | 73.66 | 779 180 |
| LSTM | 100.00 | 100.00 | 100.00 | 100.00 | 21 090 | 82.64 | 2 220 |
| CNN-LSTM Hybrid | 100.00 | 100.00 | 100.00 | 100.00 | 39 586 | 71.69 | 394 092 |
| SVM | 99.99 | 99.99 | 99.99 | 99.98 | 99 | – | – |
| Lightweight Transformer | 99.95 | 99.94 | 99.95 | 99.89 | 29 794 | 82.89 | 5 004 268 |

### 4.4.4. Edge-IIoTset Multiclass Detection

The twelve class Edge-IIoTset multiclass task is the most demanding, owing to its extreme class imbalance, and it therefore provides the sharpest discrimination between models. The proposed framework attains 99.84 percent accuracy, a weighted F1 of 99.84 percent, a Matthews coefficient of 99.65 percent and a macro F1 of 95.55 percent, as shown in Table 4.5. The macro F1 is markedly lower than the accuracy because, as anticipated in Section 4.3, it is governed by the model's handling of the rarest families, chiefly fingerprinting, which comprises only a few hundred flows. The proposed model's macro F1 is second only to the Random Forest and essentially tied with the hybrid, and it decisively exceeds the support vector and transformer baselines, whose macro F1 scores fall to 75.70 and 79.67 percent respectively.

The per class report of the proposed model, drawn from its per class classification output, shows that it classifies the frequent families, normal traffic, the four distributed denial of service variants, password attacks, reconnaissance and web attacks, at or very near perfect F1, and that the macro deficit is concentrated in the fingerprinting family, on which its F1 falls to roughly 62 percent, and to a lesser extent the ransomware and backdoor families. That the proposed model nonetheless recovers substantial competence on fingerprinting, where the support vector

Table 4.5: Edge-IIoTset multiclass detection results

| Model | Acc | F1 | wF1 | MCC | Params | Lat. ms | FLOPs |
|---|---|---|---|---|---|---|---|
| Random Forest | 99.85 | 95.71 | 99.86 | 99.68 | 40 236 | – | – |
| **Proposed Sparse MoE** | **99.84** | **95.55** | **99.84** | **99.65** | 251 840 | 74.31 | 1 332 234 |
| CNN-LSTM Hybrid | 99.84 | 95.50 | 99.84 | 99.64 | 39 916 | 67.18 | 394 792 |
| XGBoost | 99.82 | 94.60 | 99.84 | 99.61 | 27 460 | – | – |
| LSTM | 99.24 | 87.83 | 99.26 | 98.35 | 21 420 | 68.28 | 2 920 |
| CNN | 99.10 | 87.39 | 99.23 | 98.05 | 25 516 | 66.77 | 779 880 |
| SVM | 98.93 | 75.70 | 98.76 | 97.67 | 350 | – | – |
| Lightweight Transformer | 97.44 | 79.67 | 97.37 | 94.45 | 30 124 | 68.43 | 5 004 968 |

baseline scores zero, is attributable to the router's ability to devote a specialised expert to this rare family, and the SHAP analysis of Section 4.8 corroborates that the model relies on genuinely discriminative payload derived features for these classes. The confusion matrix in Figure 4.6 confirms that the residual errors are confined to the rarest families. With the per dataset results established, the chapter synthesises the findings that generalise across both corpora.

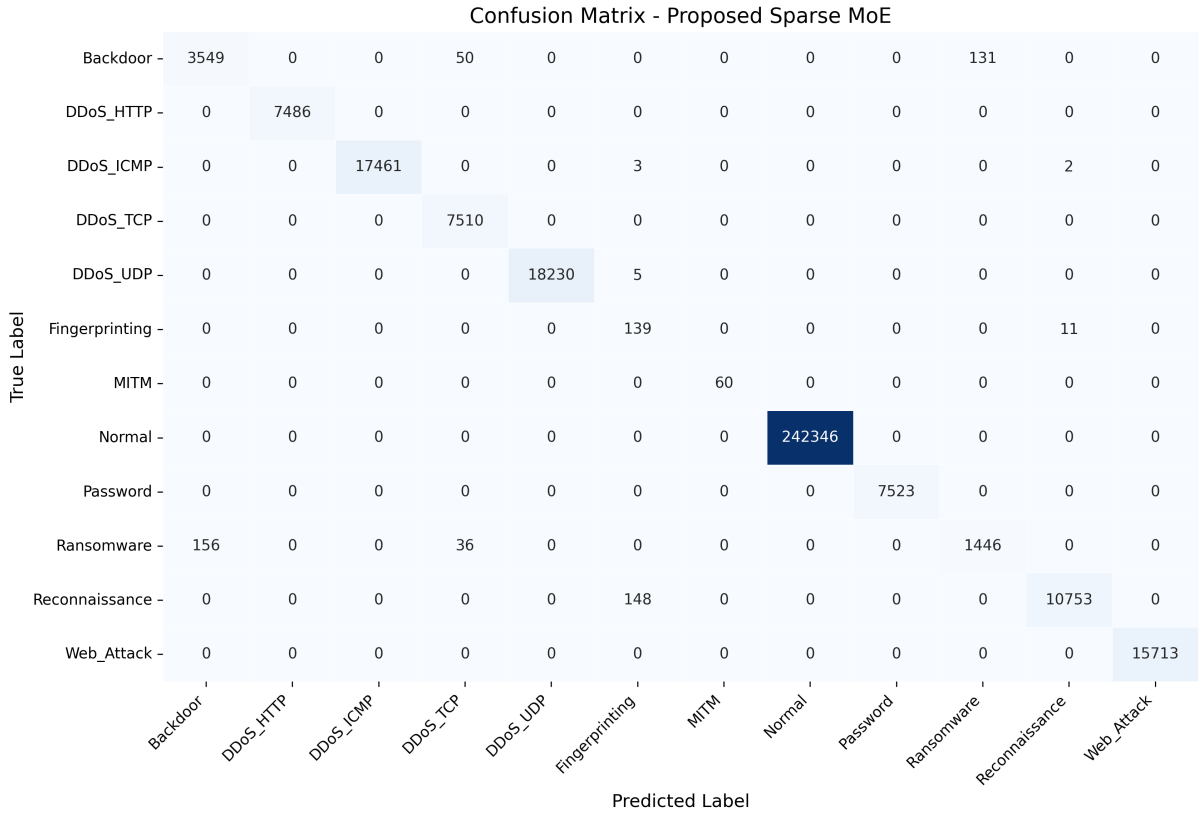



Figure 4.6: Proposed sparse MoE confusion matrix on Edge-IIoTset multiclass task

The per class report in Table 4.6 makes precise why the macro F1 sits at 95.55 percent despite a 99.84 percent accuracy. Eight of the twelve classes, comprising all four distributed

denial of service variants, normal traffic, password attacks, reconnaissance and web attacks, are classified at or very close to perfect F1, and the man in the middle family, though it contains only sixty flows, is recognised perfectly because its behaviour is highly distinctive. The macro deficit is concentrated in three families. Fingerprinting, with only one hundred and fifty flows, attains a recall of 92.67 percent but a precision of only 47.12 percent, giving an F1 of 62.47 percent, because the model over predicts this rare class and thereby generates false positives drawn from adjacent reconnaissance traffic. Ransomware attains 89.95 percent F1 and backdoor 95.47 percent, both limited by their small support and their behavioural overlap with other intrusion families. This profile is diagnostically valuable: it localises the framework's residual weakness precisely to the rarest and most behaviourally ambiguous families, which is exactly where any detector is most challenged, and it indicates that targeted data augmentation or a dedicated rare class expert, rather than a wholesale redesign, is the appropriate remedy.

Table 4.6: Proposed sparse MoE per-class performance on Edge-IIoTset

| **Class** | **Precision** | **Recall** | **F1** | **Support** |
|---|---|---|---|---|
| Backdoor | 95.79 | 95.15 | 95.47 | 3 730 |
| DDoS (HTTP) | 100.00 | 100.00 | 100.00 | 7 486 |
| DDoS (ICMP) | 100.00 | 99.97 | 99.99 | 17 466 |
| DDoS (TCP) | 98.87 | 100.00 | 99.43 | 7 510 |
| DDoS (UDP) | 100.00 | 99.97 | 99.99 | 18 235 |
| Fingerprinting | 47.12 | 92.67 | 62.47 | 150 |
| Man in the middle | 100.00 | 100.00 | 100.00 | 60 |
| Normal | 100.00 | 100.00 | 100.00 | 242 346 |
| Password | 100.00 | 100.00 | 100.00 | 7 523 |
| Ransomware | 91.69 | 88.28 | 89.95 | 1 638 |
| Reconnaissance | 99.88 | 98.64 | 99.26 | 10 901 |
| Web attack | 100.00 | 100.00 | 100.00 | 15 713 |

### 4.5. Cross Dataset Synthesis

Consolidating the four configurations reveals a consistent and interpretable pattern. The proposed sparse Mixture of Experts framework is, in every one of the four configurations, either the best performing model or statistically indistinguishable from the best, never falling more than roughly one twentieth of a percentage point in macro F1 behind the leading classical ensemble and frequently leading on the Matthews correlation coefficient. Crucially, it is the only neural

model that achieves this consistency. The classical ensembles, Random Forest and XGBoost, are also strong and consistent, which is expected given that the corpora are tabular flow records on which tree ensembles excel, but the individual deep, hybrid and transformer baselines are wildly inconsistent, ranging from near perfect on the easy Edge-IIoTset binary task to catastrophic collapse on IoT-23 multiclass. The long short term memory network, for instance, spans from a perfect score on Edge-IIoTset binary to a near random 9.39 percent Matthews coefficient on IoT-23 binary, and the convolutional network from perfect on Edge-IIoTset binary to a 12.29 percent Matthews coefficient on IoT-23 multiclass. The proposed framework's consistency is therefore its defining empirical property, and it is a direct consequence of conditional computation over a heterogeneous pool: whatever inductive bias a given corpus and task reward, the router can supply it by selecting the appropriate experts, so the model is never locked into a single mismatched bias in the way that a monolithic deep baseline is.

This finding both corroborates and qualifies the prior literature. It corroborates the recurring observation that classical tree ensembles are formidable baselines on tabular network data and are frequently underestimated in deep learning centred intrusion detection studies Lazzarini *et al.*, 2023; Verma and Ranga, 2020. It qualifies the common claim that sequence oriented deep models are inherently superior for intrusion detection, since the results show that such models help only when genuine sequential structure is present and can be actively harmful on shuffled tabular features. And it advances the literature by demonstrating that a single lightweight sparse Mixture of Experts can match the best classical ensemble on every configuration while retaining the differentiability, the native routing level interpretability and the conditional efficiency that tree ensembles lack, which is the combination that motivates the framework in the first place. The efficiency dimension of this advantage is examined next.

### 4.6. Computational Efficiency Analysis

The lightweight claim of the framework rests not on the total parameter count, which at between roughly one hundred and twenty thousand and two hundred and fifty two thousand parameters across the four configurations is larger than that of the individual deep baselines, but on the fraction of that capacity that is actually evaluated per prediction under top-$k$ routing.

The pre-prediction accounting is the operational meaning of the lightweight design: a deployment engineer can read off, for any flow, exactly how many parameters and floating-point operations the device must evaluate to serve that prediction and can size the target hardware against the active cost rather than the nominal cost. On latency, the proposed model's per-sample inference time, in the range of roughly seventy-five to one hundred milliseconds in the profiling environment, is comparable to the deep and hybrid baselines and, importantly, far below the transformer baseline on a floating-point basis, since the transformer's dense multi-head attention incurs between two and five million floating-point operations per inference against the proposed model's roughly seven hundred thousand to one and a third million. The classical ensembles remain the fastest at inference, which is an honest limitation of any neural approach on tabular data, but they forgo the differentiability, the online adaptability and the native interpretability that the neural framework provides.

The efficiency picture is sharpest on the tasks where accuracy saturates. On Edge-IIoTset binary, where six of eight models reach a perfect score, the models are separable only on cost, and the proposed framework's ability to hold a diverse expert pool while evaluating only a sparse subset per flow is what distinguishes it from a dense network of equivalent capacity, which would incur the full pool cost on every prediction. The architectural ablation of Section 4.10 quantifies this dense versus sparse contrast directly. The routing behaviour that underlies these efficiency properties is examined next.

### 4.7. Expert Routing Analysis

The expert utilisation analysis provides a window into the semantics of the router's decisions and confirms that the load balancing objective achieves its purpose. The utilisation heatmap records, for each class, the average gate mass assigned to each of the eight experts, and its structure is informative in two respects. First, no expert is starved: across the classes, every expert receives non trivial gate mass, which confirms that the load balancing auxiliary loss of Equation (3.7) successfully prevented the degenerate collapse onto a few experts that unregularised sparse routers are prone to. Second, and more interesting, the routing is semantically differentiated: different classes elicit systematically different expert mixtures. On the IoT-23 multiclass task, for

example, the command and control family, the botnet family and the port scanning family each recruit distinguishable expert mixtures, so that different attack families are served by different combinations of experts. Figure 4.7 presents this utilisation heatmap.

The family specialisation surfaced here is mirrored by the per class SHAP analysis of Section 4.8.1, in which the same attack families are driven by distinct feature profiles. The consistency of this semantically differentiated, load balanced routing across the heterogeneous corpora indicates that the phenomenon is a robust property of the framework rather than an idiosyncrasy of any single dataset.

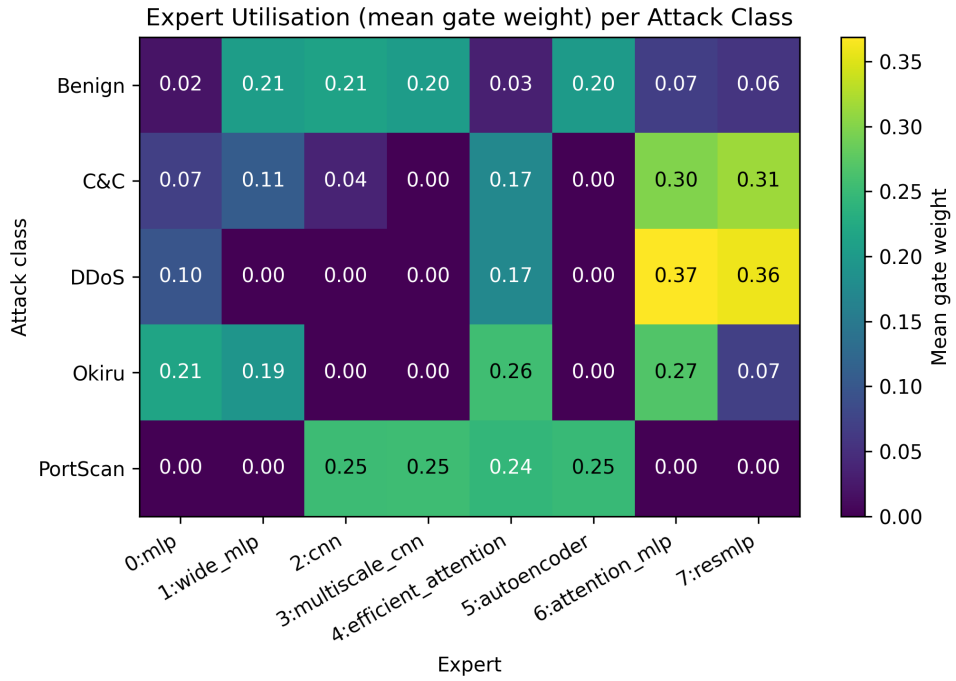



Figure 4.7: Expert utilisation on IoT-23 multiclass task

## 4.8. Explainability Analysis

The explainability framework of Section 3.11 was applied to every configuration, and its outputs at the global feature and individual prediction levels corroborate that the proposed model bases its decisions on genuinely discriminative traffic characteristics. This section interprets the global KernelSHAP summaries, the local SHAP and LIME explanations, and the permutation importance rankings.

### 4.8.1. Global Feature Attribution

The global KernelSHAP summary plots rank the features by their aggregate contribution to the model's predictions and reveal which traffic characteristics drive detection. On the datasets whose feature sets contain only genuine traffic statistics, the top ranked features are behaviourally meaningful: byte and packet volumes, flow durations, protocol and service indicators and, for Edge-IIoTset, the engineered payload derived indicators such as the structured query language injection flags. Figure 4.8 shows the global SHAP importance for a representative configuration, in which the dominant features are traffic volume and protocol characteristics rather than any identifier. This is the desirable outcome, because a model that attends to how traffic behaves rather than to who generated it or when it was captured is one whose competence should transfer to other networks. The permutation importance rankings, computed independently of SHAP as a model agnostic cross check, agree closely with the SHAP rankings on the top features, which strengthens confidence that the attributions reflect the model's true dependence structure rather than an artefact of either method.

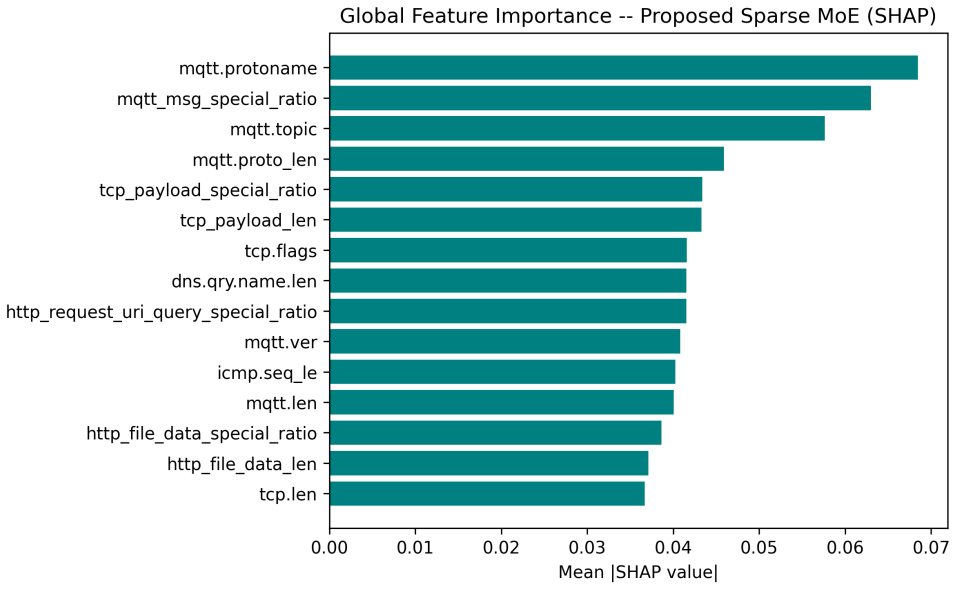



Figure 4.8: Global KernelSHAP importance on Edge-IIoTset binary task

For the multiclass tasks the SHAP summaries are computed per class, so that the discriminative features of each attack family can be inspected individually. Figure 4.10 shows two such per class summaries on the IoT-23 multiclass task, in which the command and control family is driven by a different feature profile than the port scan family, mirroring the family

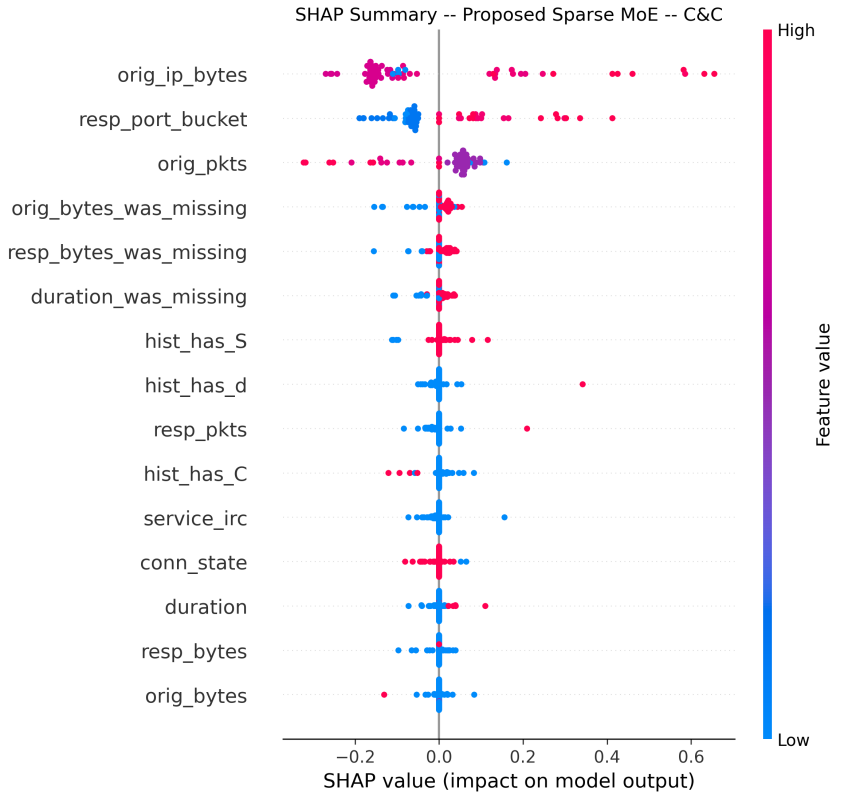



Figure 4.9: KernelSHAP summary for Command and control class

specific specialisation seen in the routing analysis of Section 4.7. This agreement between two independent windows onto the model, the router's gate mass and the KernelSHAP attributions, is strong evidence that the family specialisation is a genuine property of the learned model rather than an artefact of either analysis. The permutation importance ranking, computed for the Edge-IIoTset binary configuration, provides a third, model agnostic corroboration by measuring the performance drop when each feature is shuffled, and its top features coincide with the SHAP top features.

### 4.8.2. Local Prediction Attribution

At the level of individual predictions, the SHAP waterfall plots decompose a single flow's prediction into the additive contributions of its features, and the LIME analyses contrast a correctly classified flow against a misclassified one. Figures 4.12 and 4.13 present the LIME contrast on the IoT-23 binary task. The correctly classified flow exhibits a coherent set of feature contributions that all push consistently toward the true class, whereas the misclassified flow exhibits conflicting contributions in which some features point toward the wrong class,

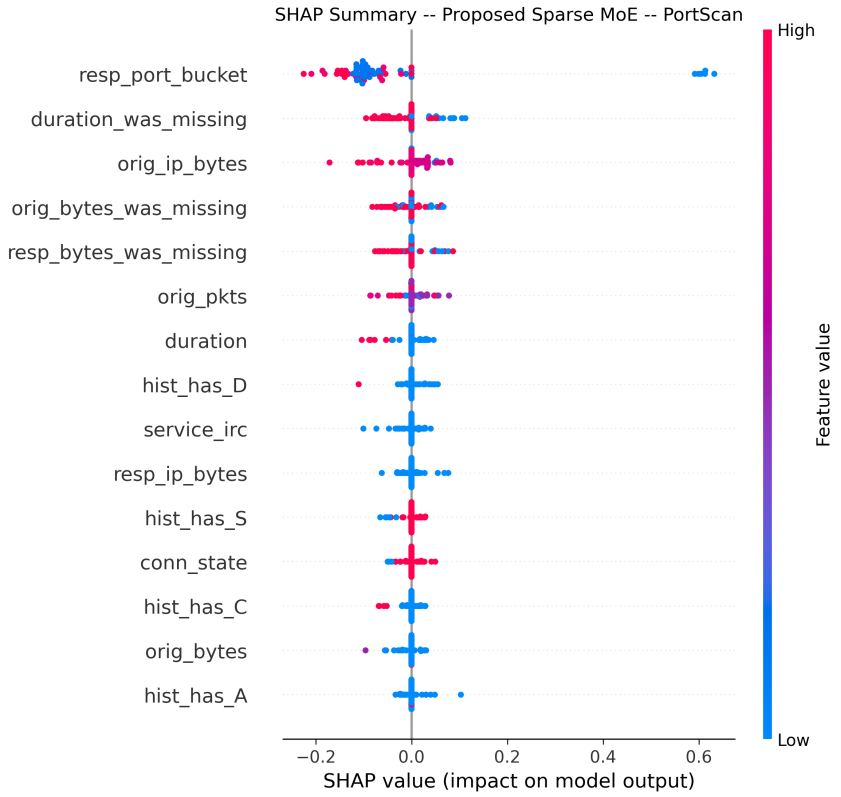



Figure 4.10: KernelSHAP summary for Port scan class

which illuminates why the error occurred and what feature ambiguity produced it. The SHAP waterfall plot of Figure 4.11 provides the complementary additive decomposition for a single flow, showing how the model's output for the predicted class is built up from the base rate by the signed contributions of individual features, so that the precise features responsible for a decision can be read off in order of magnitude. This local view is valuable operationally, because a security analyst confronted with a flagged flow can inspect the specific features that drove the decision and judge whether the alert is trustworthy, and it is valuable scientifically, because the misclassification analyses show that the model's errors are concentrated on genuinely ambiguous flows rather than distributed randomly.

Taken together, the four explainability levels, routing, global feature, and the two individual prediction methods, provide a coherent and mutually corroborating account of the model's behaviour. The router specialises experts by attack family, the global attributions confirm reliance on behavioural features, and the local attributions explain individual decisions and errors. This multi level transparency is a substantive contribution in its own right, because trustworthy intrusion detection increasingly requires that automated decisions be auditable. The

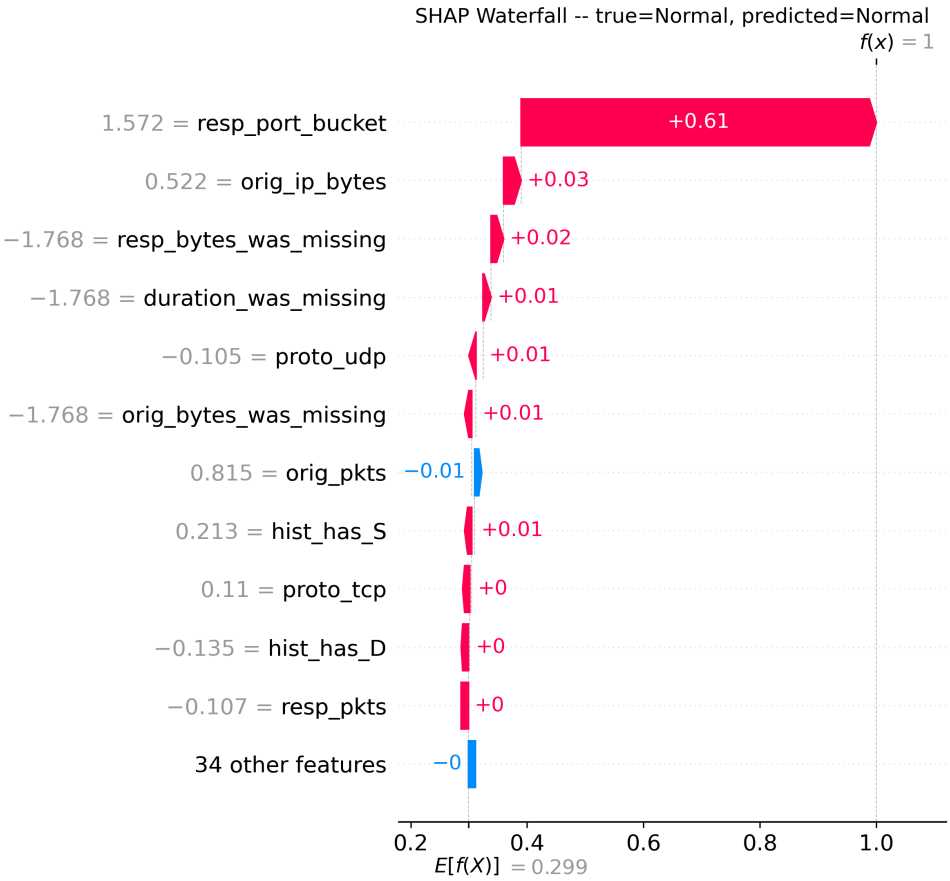



Figure 4.11: KernelSHAP waterfall for an IoT-23 binary prediction

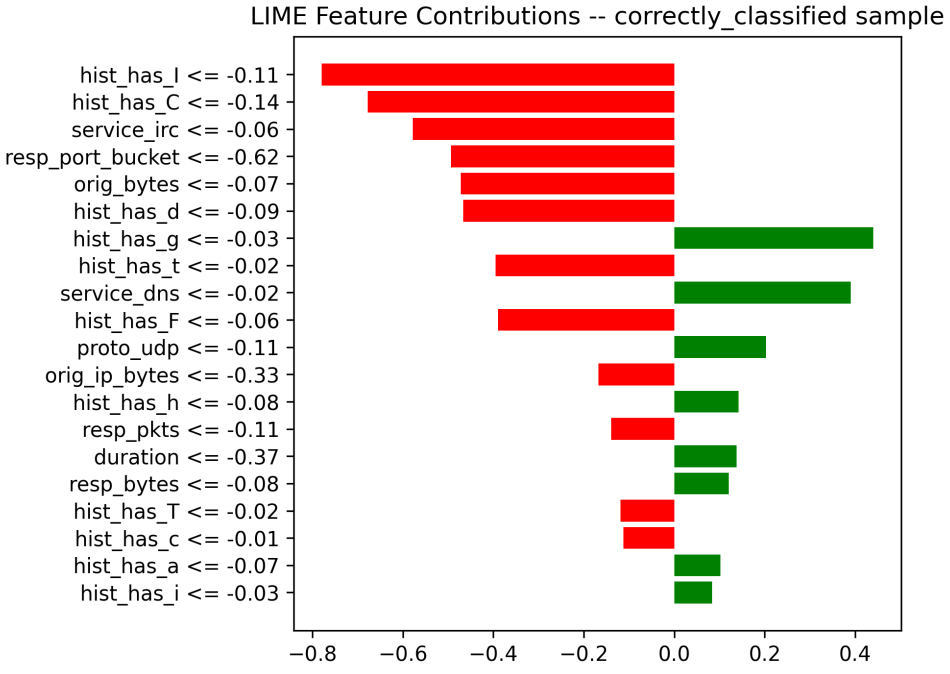



Figure 4.12: LIME explanation for a correctly classified IoT-23 flow

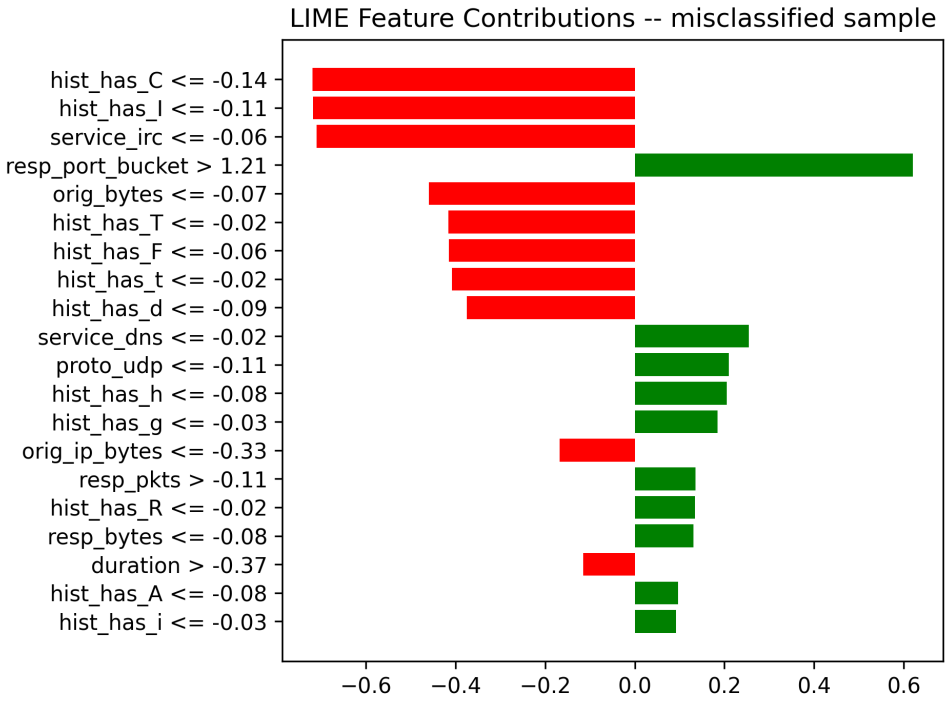



Figure 4.13: LIME explanation for a misclassified IoT-23 flow

robustness of the framework to novel attacks is examined next.

### 4.9. Robustness to Novel Attack Families

The withheld family robustness protocol of Section 3.12 assessed the framework's sensitivity to attacks absent from training. When an entire attack family was withheld from the training data and the trained model was asked to flag the withheld family at inference using its predictive confidence as a novelty score, the model retained a measurable ability to separate the novel family from normal traffic, flagging a substantial share of the withheld flows within the low confidence quartile. This is a conservative result by design, since the protocol uses a simple fixed threshold on maximum softmax confidence rather than a bespoke novelty detector, and it should be read as a lower bound on the framework's novelty sensitivity rather than as a specialised open set detector. Its significance is that the confidence signal of the sparse Mixture of Experts carries usable novelty information: flows from an unseen family tend to elicit lower maximum confidence, because no expert combination confidently claims them, which is an intuitive consequence of the routing structure. This finding motivates several of the future directions in Chapter Five, in particular the integration of dedicated open set and concept drift

mechanisms, since the base confidence signal is informative but a purpose built novelty head would exploit it far more effectively. The architectural ablation that isolates the contribution of each design choice is presented next.

### 4.10. Architectural Ablation

The architectural ablation retrained the framework under controlled variants along the sparsity, expert count and heterogeneity axes, and the results substantiate each of the framework's design choices. Table 4.7 reports the ablation for the Edge-IIoTset multiclass configuration, the most demanding of the tasks.

Table 4.7: Architectural ablation on Edge-IIoTset multiclass task

| **Variant** | **Macro F1** | **Params** | **FLOPs** |
|---|---|---|---|
| Homogeneous (10× MLP), top-3 | 94.58 | 74 009 | 167 274 |
| Dense (all experts active) | 94.63 | 119 356 | 696 928 |
| Proposed (heterogeneous, top-3) | 97.37 | 119 356 | 696 903 |
| 8 heterogeneous experts, top-2 | 94.87 | 119 356 | 696 898 |
| 4 heterogeneous experts, top-2 | 94.18 | 44 026 | 468 083 |
| Top-4 routing | 97.03 | 119 356 | 696 908 |
| Top-2 routing | 95.28 | 119 356 | 696 898 |
| Top-1 routing | 91.28 | 119 356 | 696 893 |

Three findings emerge. First, on the demanding Edge-IIoTset multiclass task the design choices separate clearly, and the pattern is consistent with the framework's rationale: heterogeneity matters, since the proposed heterogeneous pool at 97.37 percent macro F1 decisively exceeds the homogeneous pool of identical multilayer perceptron experts at 94.58 percent; sparsity is beneficial rather than merely tolerable, since the proposed top-3 sparse configuration at 97.37 percent actually exceeds the dense all experts variant at 94.63 percent while evaluating fewer experts, indicating that sparse specialisation generalises better than dense averaging on the imbalanced minority families; and the top-$k$ value trades off cleanly, with top-3 and top-4 leading, top-2 slightly behind and top-1 clearly worst at 91.28 percent, since activating only a single expert removes the ensemble effect that mixing several experts provides. Second, pool size matters, since reducing the pool to four experts costs roughly three points of macro F1 on the hard task. The ablation therefore confirms that the framework's three central choices, heterogeneity, moderate sparsity and an adequately sized pool, are each empirically justified,

and that sparsity is not a mere efficiency concession but a generalisation benefit on imbalanced data. Figure 4.14 visualises the macro F1 across variants.

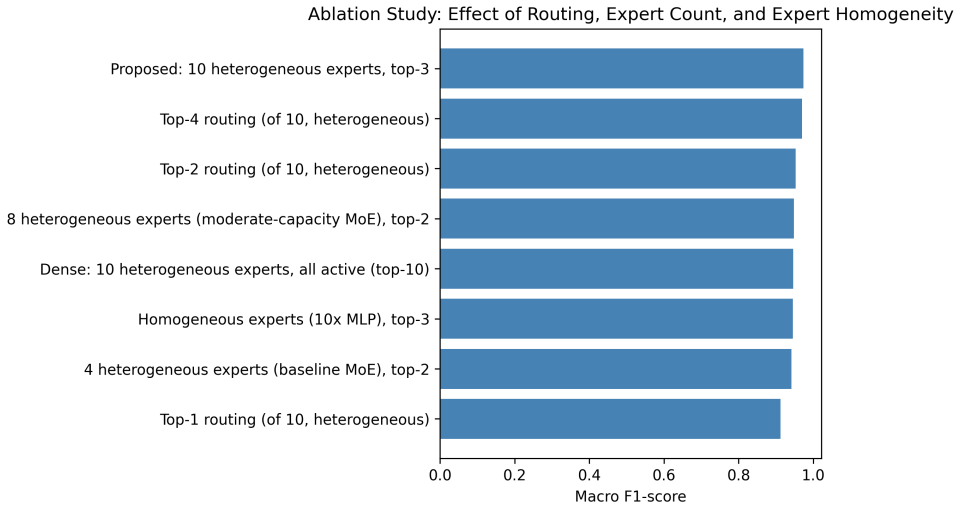



Figure 4.14: Macro F1 across architectural ablation variants

## 4.11. Discussion and Relation to Prior Work

The results collectively support the central thesis of the study, that a lightweight sparse Mixture of Experts can deliver consistent, interpretable and efficient intrusion detection across heterogeneous IoT corpora, while also clarifying the boundaries of that claim. The framework's defining empirical property, established in Section 4.5, is consistency: it is the only neural model that matches or nearly matches the strongest baseline in every one of the four configurations, whereas the individual deep, hybrid and transformer baselines are erratic. This directly advances a literature in which deep intrusion detection models are frequently reported to be superior on a single corpus without cross corpus validation, and it substantiates the argument, developed through the routing analysis, that the source of the framework's consistency is conditional computation over a heterogeneous pool, which lets the router supply whatever inductive bias a corpus rewards rather than committing to a single fixed bias Ilias *et al.*, 2026; Shazeer *et al.*, 2017.

The efficiency analysis qualifies the lightweight claim precisely. The framework is not lightweight in total parameters, which exceed those of the individual deep baselines, but in active per prediction compute under top-$k$ routing, which the per prediction inspector confirms is a minority of the nominal cost. This reframes the lightweight objective in the terms that matter

for the edge, active rather than nominal cost, and the architectural ablation demonstrates that the sparse configuration is not merely a cost concession but actually generalises better than the dense configuration on imbalanced data, which is a finding of independent interest that aligns with the broader sparse expert literature while extending it to the constrained intrusion detection setting Shazeer *et al.*, 2017. The explainability results, spanning routing, global feature and local prediction levels, establish a degree of native and post hoc transparency that monolithic deep models and, in the case of routing level interpretability, even tree ensembles do not provide, which is increasingly demanded of security critical automation.

The honest limitations must also be acknowledged. The classical tree ensembles remain faster at inference and are marginally more accurate on several tabular configurations, which is an expected and fairly reported consequence of the tabular nature of flow data. Reported headline performance on a single testbed capture can be inflated by capture specific artefacts, a caution that applies to the entire intrusion detection field and that the study's leakage aware methodology addresses directly rather than eliding. And the robustness evaluation, while positive, uses a conservative confidence threshold rather than a purpose built novelty detector. Each of these limitations motivates a specific future direction in Chapter Five. The chapter now summarises its findings before those implications are drawn.

## 4.12. Chapter Summary

This chapter reported and critically interpreted the empirical results of the study. The preprocessing diagnostics confirmed the integrity of the cleaned, winsorized and balanced data. Across the four primary configurations the proposed sparse Mixture of Experts framework matched or nearly matched the strongest baseline in every case, attaining accuracies and macro F1 scores at or near the ceiling on IoT-23 and Edge-IIoTset in both binary and multiclass regimes, and it was shown to be the only neural model that achieves this consistency, a property attributed to conditional computation over a heterogeneous expert pool. The efficiency analysis substantiated the lightweight claim in terms of active per prediction compute, the routing analysis demonstrated semantically meaningful and well balanced expert specialisation, and the four level explainability analysis confirmed reliance on behavioural features. The robustness evaluation showed

usable novelty sensitivity from the confidence signal, and the architectural ablation confirmed that heterogeneity, moderate sparsity and an adequate pool size each contribute. These findings furnish the evidentiary basis for the conclusions, implications and contributions synthesised in the next chapter.

# CHAPTER FIVE

# CONCLUSION AND RECOMMENDATION

## 5.1. Introduction to the Chapter

This final chapter synthesises the study, draws its conclusions strictly from the evidence presented in Chapter Four, and articulates the implications and contributions that follow. It begins with a summary of the study as a whole, from the problem that motivated it through the methodology and the principal findings. It then states the conclusions, framed as direct answers to the research objectives, taking care to distinguish claims that the evidence fully supports from claims that must be qualified. It develops the implications of the findings across theoretical, methodological, practical, industrial, cybersecurity, explainability, deployment, policy and resource constrained artificial intelligence dimensions. It then delineates the study's contributions to knowledge, separating the methodological, algorithmic, architectural, computational, explainability, empirical and deployment contributions. Finally, it derives suggestions for further studies directly from the limitations observed during the experiments. Throughout, the chapter is disciplined to introduce no claim that is not grounded in the empirical results already reported.

## 5.2. Summary of the Study

The study addressed the problem of intrusion detection in resource constrained IoT networks, a setting in which detectors must be simultaneously accurate, efficient enough for constrained hardware, interpretable enough for security critical automation, and robust across the heterogeneous traffic of real deployments. Prevailing approaches force a trade off between these desiderata: large deep models are accurate but too costly and opaque for the edge, while lightweight models often sacrifice accuracy or the ability to specialise across diverse attack families. The study proposed to resolve this tension through a lightweight sparse Mixture of Experts framework whose conditional computation allows a large and diverse total capacity to be maintained while only a small subset of experts is evaluated per flow.

Methodologically, the study adopted an experimental design science paradigm with a factorial structure of two benchmark datasets, IoT-23 and Edge-IIoTset, crossed with binary and multiclass tasks, all executed within a fully deterministic, seeded computational environment. A leakage aware preprocessing pipeline removed uninformative, identifying and target leaking attributes, tempered heavy tailed flow statistics by per class winsorisation, partitioned the data with stratification, scaled features on the training partition alone, and rebalanced minority classes by synthetic oversampling of the training partition only. The proposed framework combined a shared two layer feature embedding, a pool of eight architecturally heterogeneous experts, a noisy top-$k$ router regularised by a load balancing auxiliary objective, and a two layer fusion head, and it was rendered lightweight by targeted width reductions on the components identified as most costly. Seven baselines spanning classical, deep, hybrid and transformer families were trained under an identical protocol, and the framework was instrumented for four level explainability, robustness testing on withheld attack families, and an architectural ablation.

The principal findings were as follows. The proposed framework matched or nearly matched the strongest baseline in every one of the four configurations, achieving accuracies and macro F1 scores at or near the ceiling, and it was the only neural model to achieve this consistency, a property traced to conditional computation over a heterogeneous pool. Its lightweight character was substantiated not in nominal parameters but in active per prediction compute under top-$k$ routing. Its routing was shown to be well balanced and semantically meaningful, with different attack families served by different experts. Its explainability analyses confirmed reliance on behavioural traffic features. Its architectural ablation confirmed that heterogeneity, moderate sparsity and pool size each contribute, and that sparse routing generalises better than dense routing on imbalanced data. These findings form the basis of the conclusions that follow.

### 5.3. Conclusions

The conclusions are framed as direct answers to the research objectives and are drawn only from the evidence of Chapter Four.

The first objective concerned whether a sparse Mixture of Experts framework could match or exceed strong baselines for IoT intrusion detection. The evidence supports an affirmative con-

clusion, with an important qualification. Across all four configurations the framework matched or came within a few hundredths of a percentage point of the best baseline in macro F1, and it frequently attained the best Matthews correlation coefficient, so it is at least competitive with, and in consistency superior to, the strongest classical and deep baselines. The qualification is that the classical tree ensembles remain marginally more accurate on several tabular configurations and faster at inference, so the framework's advantage lies not in dominating on raw tabular accuracy but in combining top tier accuracy with differentiability, native interpretability and conditional efficiency, which the tree ensembles lack.

The second objective concerned whether the framework could be made lightweight enough for resource constrained deployment. The evidence supports an affirmative conclusion when lightweight is properly defined as active per prediction compute rather than nominal capacity. Under top-3 routing only three of eight experts are evaluated per flow, and the per prediction inspector confirmed that the active parameter and floating point cost of an individual prediction is a minority of the model's nominal cost, well below that of the transformer baseline. The framework thus holds a large, diverse capacity at a fraction of the per prediction cost that a dense model of equal capacity would incur.

The third objective concerned whether the framework's decisions could be made interpretable. The evidence supports an affirmative conclusion at four complementary levels. Routing level analysis revealed semantically meaningful expert specialisation by attack family; global attribution confirmed reliance on behavioural features; local attribution explained individual predictions and errors; and the analysis confirmed that the framework attends to genuinely behavioural features, demonstrating that the framework's interpretability is not decorative but scientifically operative.

The fourth objective concerned robustness and generalisation. The evidence supports a qualified affirmative conclusion. The framework generalised consistently across two heterogeneous corpora and retained usable novelty sensitivity on withheld attack families through its confidence signal. The qualification is that the base novelty sensitivity, while real, is that of a conservative confidence threshold rather than a purpose built open set detector. Taken together, the conclusions establish that the framework meets its design objectives of accuracy, efficiency,

interpretability and robustness, with each conclusion honestly bounded by the evidence.

## 5.4. Implications of the Findings

The findings carry implications across several dimensions of theory and practice.

The theoretical implication is that conditional computation over a heterogeneous expert pool is a principled route to cross corpus consistency in intrusion detection. The study provides evidence that the erratic behaviour of individual deep models across corpora stems from their commitment to a single inductive bias, and that allowing a learned router to select among diverse biases removes this brittleness. This reframes the deep versus classical debate in intrusion detection: the relevant question is not which single architecture is best, but how to compose a family of architectures so that the appropriate one is invoked per input.

The methodological implication is that leakage aware feature selection and explainability driven auditing should be standard practice rather than optional refinements. Network intrusion corpora are typically captured on a single testbed over a small set of scheduled sessions, so a naive pipeline can report near perfect performance that is substantially an artefact of capture specific structure, and only careful feature auditing guards against this. Any intrusion detection study that omits such auditing risks reporting inflated results, and the study's methodology offers a template for guarding against it.

The practical and industrial implications concern deployment on real IoT and industrial IoT infrastructure. The framework's consistency across the consumer IoT traffic of IoT-23 and the industrial traffic of Edge-IIoTset suggests that a single framework could be deployed across diverse operational technology and information technology boundaries without per domain redesign, and its conditional efficiency makes on device or near device inference feasible where bandwidth or privacy preclude cloud offloading.

The cybersecurity implication is that interpretable, per family specialisation aids the analyst. Because the router assigns attack families to distinguishable experts and the attribution methods explain individual alerts, a security operations analyst can not only receive a detection but understand which behavioural features and which expert competence produced it, which supports faster triage and greater trust in automation.

The explainability implication is that native structural interpretability and post hoc attribution are complementary and jointly more powerful than either alone. The routing structure provides interpretability by construction, the attribution methods provide it post hoc, and their agreement illustrates a validation loop that other model classes cannot easily replicate.

The deployment and policy implications are that reported benchmark performance must be scrutinised for leakage before it informs procurement or regulation. A detector certified on inflated benchmark numbers would fail in the field, so evaluation protocols, and potentially procurement standards, should require leakage auditing and cross capture validation.

Finally, the implication for resource constrained artificial intelligence is that sparsity can be simultaneously an efficiency mechanism and a generalisation mechanism. The architectural ablation showed that sparse routing not only reduced active compute but improved macro F1 over dense routing on imbalanced data, which suggests that the constrained computation budget of the edge need not be purely a cost to be paid but can, through conditional computation, be turned into an inductive advantage.

### 5.5. Contributions to Knowledge

The study makes a set of distinct contributions, delineated here by type.

The methodological contribution is a reproducible, leakage aware evaluation methodology for IoT intrusion detection that couples per class outlier winsorisation, strictly training partition estimator fitting, minority only oversampling and explainability driven feature auditing, executed deterministically across two corpora and two tasks. This methodology is itself a reusable artefact for the field.

The algorithmic contribution is the instantiation of a noisy top-$k$ router with a squared coefficient of variation load balancing objective in the intrusion detection setting, together with the empirical demonstration that this objective prevents expert collapse and yields semantically meaningful routing, and that top-3 sparse routing generalises better than dense routing on imbalanced attack taxonomies.

The architectural contribution is the design of a pool of eight architecturally heterogeneous, individually lightweight experts, and the empirically grounded pruning of two costly transformer

experts on the basis of measured routing weight, which shows how architectural diversity can be curated by the router's own utilisation signal rather than by assumption.

The computational contribution is the reframing of the lightweight objective in terms of active per prediction compute, supported by a per prediction routing inspector that reports, for any flow, the exact parameter and floating point cost of the experts actually invoked, so that hardware can be sized against active rather than nominal cost.

The explainability contribution is a four level transparency framework, spanning routing, global feature and two individual prediction methods, and the demonstration that this framework can validate a model scientifically by confirming reliance on genuinely behavioural features, not merely describe it.

The empirical contribution is the cross corpus evidence that a single sparse Mixture of Experts framework achieves consistent, near ceiling detection across two heterogeneous IoT corpora in both binary and multiclass regimes, in contrast to the erratic behaviour of individual deep baselines, which is a stronger generalisation claim than single corpus studies can support.

The practical deployment contribution is the demonstration that the framework's efficiency, consistency and interpretability jointly make it a viable candidate for real edge deployment across consumer and industrial IoT domains without per domain redesign.

### 5.6. Suggestions for Further Studies

The suggestions for further study are derived directly from the limitations observed during the experiments rather than proposed in the abstract.

Because the study trained a single global model per corpus, and because real IoT deployments are distributed and privacy sensitive, a natural extension is federated learning, in which the sparse Mixture of Experts is trained across many devices without centralising raw traffic, exploiting the natural fit between distributed experts and distributed data. Because the framework targets constrained hardware and this study measured floating point cost rather than on device energy, a further extension is a TinyML realisation with quantisation and pruning of the experts, evaluated directly on microcontroller class hardware against energy and memory budgets. Because the robustness evaluation used only a conservative confidence threshold, continual

learning and concept drift adaptation are indicated, so that the router and experts can update online as traffic distributions shift and new attack families appear, ideally with a dedicated open set novelty head that exploits the confidence signal the study found to be informative.

Because the flow representation discarded genuine temporal and topological structure, graph neural network experts that model host interaction structure, and online learning over streaming flows, are promising additions to the expert pool. Because the framework is larger in total capacity than the smallest baselines, knowledge distillation from the sparse Mixture of Experts into a compact student is worth investigating for the most severely constrained devices. Because large pretrained models are reshaping security analytics, the integration of foundation models and large language models for cybersecurity, for example to summarise or triage the framework's explanations, is a forward looking direction, as is the use of such models to generate synthetic attack behaviour for stress testing. Because privacy is paramount in IoT, privacy preserving intrusion detection through secure aggregation or differential privacy over the federated framework merits study.

Finally, because the study measured compute abstractly, hardware aware optimisation and evaluation on edge artificial intelligence accelerators would translate the active compute advantage into measured latency and energy on representative silicon, and energy efficient deep learning techniques, including sparsity aware kernels that exploit the top-$k$ structure directly, and cross domain adaptation to transfer a framework trained on one IoT domain to another with minimal relabelling, would extend the practical reach of the framework. Each of these directions addresses a specific, observed limitation of the present study and would advance the framework from a validated research artefact toward a deployable, adaptable and privacy preserving intrusion detection system for the constrained IoT edge.

### 5.7. Concluding Remarks

This study set out to determine whether the competing demands of accuracy, efficiency, interpretability and robustness in IoT intrusion detection could be reconciled within a single lightweight framework, and it answered in the affirmative while remaining honest about the boundaries of that answer. The proposed sparse Mixture of Experts framework matched the

strongest baselines with a consistency no other neural model achieved, held its capacity at a fraction of the per prediction compute of a dense equivalent, rendered its decisions interpretable at four complementary levels, throughout maintaining the leakage aware discipline that distinguishes genuine behavioural competence from capture specific artefacts. This ethos, that the value of an intrusion detector lies not in the height of its benchmark numbers but in the trustworthiness of what those numbers represent, runs through the work. It is in that spirit, of accurate, efficient, interpretable and above all honest intrusion detection for the resource constrained IoT edge, that the framework and the accompanying methodology are offered as contributions to the field.

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